RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETORESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO
2021 Revised MARPOL Annex VI
THE MARINE ENVIRONMENT PROTECTION COMMITTEE,
RECALLING Article 38(a) of the Convention on the International Maritime Organization concerning the functions of the Marine Environment Protection Committee (the Committee) conferred upon it by international conventions for the prevention and control of marine pollution from ships,
RECALLING ALSO article 16 of the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocols of 1978 and 1997 relating thereto (MARPOL), which specifies the amendment procedure and confers upon the appropriate body of the Organization the function of considering amendments thereto for adoption by the Parties,

which specifies the amendment procedure and confers upon the appropriate body of the Organization the function of considering amendments thereto for adoption by the Parties,
RECALLING FURTHER that the Committee, at its seventy-second session, adopted resolution MEPC.304(72) on the Initial IMO Strategy on reduction of GHG emissions from ships,
HAVING CONSIDERED, at its seventy -sixth session, proposed amendments to MARPOL Annex VI concerning mandatory goal-based technical and operational measures to reduce carbon intensity of international shipping and exemption of unmanned non-self-propelled (UNSP) barges from certain survey and certification requirements, which were circulated in accordance with article 16(2)(a) of MARPOL,
HAVING ALSO CONSIDERED, at its seventy-sixth session, the comprehensive assessment of the impacts of the proposed amendments to MARPOL Annex VI on States, including on developing countries, especially on least developed countries (LDCs) and small island developing States (SIDS),
1 ADOPTS, in accordance with article 16(2)(d) of MARPOL, the amendments to MARPOL Annex VI, the text of which is set out in the annex to the present resolution;

developing States (SIDS),
1 ADOPTS, in accordance with article 16(2)(d) of MARPOL, the amendments to MARPOL Annex VI, the text of which is set out in the annex to the present resolution;
2 DETERMINES, in accord ance with article 16(2)(f)(iii) of MARPOL, that the amendments to MARPOL Annex VI shall be deemed to have been accepted on 1 May 2022 unless prior to that date not less than one third of the Parties or Parties the combined merchant fleets of which constitute not less than 50% of the gross tonnage of the world's merchant fleet have communicated to the Organization their objection to the amendments;
3 INVITES the Parties to note that, in accordance with article 16(2)(g)(ii) of MARPOL, the amendments to MARPO L Annex VI shall enter into force on 1 November 2022 upon its acceptance in accordance with paragraph 2 above;
4 INVITES ALSO the Parties to consider and initiate as soon as possible the development of a Carbon Intensity Code;
5 INVITES the Organization, mindful of the review clauses provided for in regulations 25.3 and 28.11 of the amendments to MARPOL Annex VI, to initiate the respective reviews as early as possible;

5 INVITES the Organization, mindful of the review clauses provided for in regulations 25.3 and 28.11 of the amendments to MARPOL Annex VI, to initiate the respective reviews as early as possible;

6 INVITES ALSO the Organization to keep under review the impacts on States of the aforesaid amendments to MARPOL Annex VI, paying particular attention to the needs of developing countries, especially LDCs and SIDS, so that any necessary adjustments can be made;

7 AGREES to undertake a lessons -learned exercise from the comprehensive impact assessment of the amendments to MARPOL Annex VI, with a view to improving the procedure for conducting future impact assessments taking into account the Procedure for assessing impacts on States of candidate measures (MEPC.1/Circ.885) and the terms of reference for the impact assessment of the short-term measure;

8 ENCOURAGES the Parties to consider early application of the aforesaid amendments;

impacts on States of candidate measures (MEPC.1/Circ.885) and the terms of reference for the impact assessment of the short-term measure;

8 ENCOURAGES the Parties to consider early application of the aforesaid amendments;

9 REQUESTS the Secretary-General, for the purposes of article 16(2)(e) of MARPOL, to transmit certified copies of the present resolution and the text of the amendments to MARPOL Annex VI contained in the annex to all Parties to MARPOL;

10 REQUESTS ALSO the Secretary-General to transmit copies of the present resolution and its annex to Members of the Organization which are not Parties to MARPOL.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 3 -

ANNEX

MARPOL ANNEX VI

REGULATIONS FOR THE PREVENTION OF AIR POLLUTION FROM SHIPS

Chapter 1 General

Regulation 1
Application

The provisions of this Annex shall apply to all ships, except where expressly provided otherwise.

Regulation 2
Definitions

1 For the purpose of this Annex:

Chapter 1 General

Regulation 1
Application

The provisions of this Annex shall apply to all ships, except where expressly provided otherwise.

Regulation 2
Definitions

1 For the purpose of this Annex:

.1 Annex means Annex VI to the International Convention for the Prevention of Pollution from Ships, 1973 (MARPOL), as modified by the Protocol of 1978 relating thereto, and as modified by the Protocol of 1997, as amended by the Organization, provided that such amendments are adopted and brought into force in accordance with the provisions of article 16 of the present Convention.

.2 A similar stage of construction means the stage at which:

.1 construction identifiable with a specific ship begins; and

.2 assembly of that ship has commenced comprising at least 50 tonnes or one per cent of the estimated mass of all structural material, whichever is less.

.3 Anniversary date means the day and the month of each year that will correspond to the date of expiry of the International Air Pollution Prevention Certificate.

.4 Audit means a systematic, independent and documented process for

.3 Anniversary date means the day and the month of each year that will correspond to the date of expiry of the International Air Pollution Prevention Certificate.

.4 Audit means a systematic, independent and documented process for obtaining audit evidence and evaluating it objectively to determine the extent to which audit criteria are fulfilled.

.5 Audit Scheme means the IMO Member State Audit Scheme established by the Organization and taking into account the guidelines developed by the Organization.

.6 Audit Standard means the Code for Implementation.

.7 Auxiliary control device means a system, function or control strategy installed on a marine diesel engine that is used to protect the engine and/or its ancillary equipment against operating conditions that could result in damage or failure, or that is used to facilitate the starting of the engine. An auxiliary control device may also be a strategy or measure that has been satisfactorily demonstrated not to be a defeat device.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS,

not to be a defeat device.

.8 Code for Implementation means the IMO Instruments Implementation Code (III Code) adopted by the Organization by resolution A.1070(28).

.9 Continuous feeding is defined as the process whereby waste is fed into a combustion chamber without human assistance while the incinerator is in normal operating conditions with the combustion chamber operative temperature between 850°C and 1,200°C.

.10 Defeat device means a device that measures, senses or responds to operating variables (e.g. engine speed, temperature, intake pressure or any other parameter) for the purpose of activating, modulating, delaying or deactivating the operation of any component or the function of the emission control system such that the effectiveness of the emission control system is reduced under conditions encountered during normal operation, unless the use of such a device is substantially included in the applied emission certification

control system such that the effectiveness of the emission control system is reduced under conditions encountered during normal operation, unless the use of such a device is substantially included in the applied emission certification test procedures.

.11 Electronic Record Book means a device or system, approved by the Administration, used to electronically record the required entries for discharges, transfers and other operations as required under this Annex in lieu of a hard copy record book.

.12 Emission means any release of substances, subject to control by this Annex, from ships into the atmosphere or sea.

.13 Emission control area means an area where the adoption of special mandatory measures for emissions from ships is required to prevent, reduce and control air pollution from NO x or SOx and particulate matter or all three types of emissions and their attendant adverse impacts on human health and the environment. Emission control areas shall include those listed in, or designated under, regulations 13 and 14 of this Annex.

.14 Fuel oil means any fuel delivered to and intended for combustion purposes for

the environment. Emission control areas shall include those listed in, or designated under, regulations 13 and 14 of this Annex.

.14 Fuel oil means any fuel delivered to and intended for combustion purposes for propulsion or operation on board a ship, including gas, distillate and residual fuels.

.15 Gross tonnage means the gross tonnage calculated in accordance with the tonnage measurement regulations contained in Annex I to the International Convention on Tonnage Measurements of Ships, 1969, or any successor Convention.

.16 In-use sample means a sample of fuel oil in use on a ship.

.17 Installations in relation to regulation 12 of this Annex means the installation of systems, equipment, including portable fire-extinguishing units, insulation, or other material on a ship, but excludes the repair or recharge of previously installed systems, equipment, insulation or other material, or the recharge of portable fire-extinguishing units.

.18 Installed means a marine diesel engine that is or is intended to be fitted on a ship, including a portable auxiliary marine diesel engine, only if its fuelling,

portable fire-extinguishing units.

.18 Installed means a marine diesel engine that is or is intended to be fitted on a ship, including a portable auxiliary marine diesel engine, only if its fuelling, cooling or exhaust system is an integral part of the ship. A fuelling system is considered integral to the ship only if it is permanently affixed to the ship. This definition includes a marine diesel engine that is used to supplement or augment the installed power capacity of the ship and is intended to be an integral part of the ship.

.19 Irrational emission control strategy means any strategy or measure that, when the ship is operated under normal conditions of use, reduces the effectiveness of an emission control system to a level below that expected on the applicable emission test procedures.

.20 Low-flashpoint fuel means gaseous or liquid fuel oil having a flashpoint lower

of an emission control system to a level below that expected on the applicable emission test procedures.

.20 Low-flashpoint fuel means gaseous or liquid fuel oil having a flashpoint lower than otherwise permitted under paragraph 2.1.1 of regulation 4 of chapter II-2 of the International Convention for the Safety of Life at Sea (SOLAS), 1974, as amended.

.21 Marine diesel engine means any reciprocating internal combustion engine operating on liquid or dual fuel, to which regulation 13 of this Annex applies, including booster/compound systems if applied. In addition, a gas-fuelled engine installed on a ship constructed on or after 1 March 2016 or a gas-fuelled additional or non-identical replacement engine installed on or after that date is also considered as a marine diesel engine.

.22 MARPOL delivered sample means the sample of fuel oil delivered in accordance with regulation 18.8.1 of this Annex.

.23 NOx Technical Code means the Technical Code on Control of Emission of Nitrogen Oxides from Marine Diesel Engines adopted by resolution 2 of the 1997 MARPOL Conference, as amended by the Organization, provided

.23 NOx Technical Code means the Technical Code on Control of Emission of Nitrogen Oxides from Marine Diesel Engines adopted by resolution 2 of the 1997 MARPOL Conference, as amended by the Organization, provided that such amendments are adopted and brought into force in accordance with the provisions of article 16 of the present Convention.

.24 Onboard sample means a sample of fuel oil intended to be used or carried for use on board that ship.

.25 Ozone-depleting substances means controlled substances defined in paragraph (4) of article 1 of the Montreal Protocol on Substances that Deplete the Ozone Layer, 1987, listed in Annexes A, B, C or E to the said Protocol in force at the time of application or interpretation of this Annex.

Ozone-depleting substances that may be found on board ship include, but are not limited to:

Halon 1211 Bromochlorodifluoromethane
Halon 1301 Bromotrifluoromethane
Halon 2402 1,2-Dibromo-1,1,2,2-tetrafluoroethane (also known as Halon 114B2)
CFC-11 Trichlorofluoromethane
CFC-12 Dichlorodifluoromethane
CFC-113 1,1,2-Trichloro-1,2,2-trifluoroethane
CFC-114 1,2-Dichloro-1,1,2,2-tetrafluoroethane

Halon 2402 1,2-Dibromo-1,1,2,2-tetrafluoroethane (also known as Halon 114B2)
CFC-11 Trichlorofluoromethane
CFC-12 Dichlorodifluoromethane
CFC-113 1,1,2-Trichloro-1,2,2-trifluoroethane
CFC-114 1,2-Dichloro-1,1,2,2-tetrafluoroethane
CFC-115 Chloropentafluoroethane

.26 Shipboard incineration means the incineration of wastes or other matter on board a ship, if such wastes or other matter were generated during the normal operation of that ship.

.27 Shipboard incinerator means a shipboard facility designed for the primary purpose of incineration.

.28 Ships constructed means ships the keels of which are laid or that are at a similar stage of construction.

.29 Sludge oil means sludge from the fuel oil or lubricating oil separators, waste lubricating oil from main or auxiliary machinery, or waste oil from bilge water separators, oil filtering equipment or drip trays.

.29 Sludge oil means sludge from the fuel oil or lubricating oil separators, waste lubricating oil from main or auxiliary machinery, or waste oil from bilge water separators, oil filtering equipment or drip trays.

.30 Sulphur content of fuel oil means the concentration of sulphur in a fuel oil, measured in % m/m as tested in accordance with a standard acceptable to the Organization.

.31 Tanker in relation to regulation 15 of this Annex means an oil tanker as defined in regulation 1 of Annex I to the present Convention or a chemical tanker as defined in regulation 1 of Annex II to the present Convention.

.32 Unmanned non-self-propelled (UNSP) barge means a barge that:

.1 is not propelled by mechanical means;

.2 has no system, equipment and/or machinery fitted that may generate emissions regulated by this Annex; and

.3 has neither persons nor living animals on board.

2 For the purpose of chapter 4:

.1 A ship delivered on or after 1 September 2019 means a ship:

.1 for which the building contract is placed on or after 1 September 2015; or

.2 in the absence of a building contract, the keel of which is laid, or which

.1 A ship delivered on or after 1 September 2019 means a ship:

.1 for which the building contract is placed on or after 1 September 2015; or

.2 in the absence of a building contract, the keel of which is laid, or which is at a similar stage of construction, on or after 1 March 2016; or

.3 the delivery of which is on or after 1 September 2019.

.2 Attained annual operational CII is the operational carbon intensity indicator value achieved by an individual ship in accordance with regulations 26 and 28 of this Annex.

.3 Attained EEDI is the EEDI value achieved by an individual ship in accordance with regulation 22 of this Annex.

.4 Attained EEXI is the EEXI value achieved by an individual ship in accordance with regulation 23 of this Annex.

.5 Bulk carrier means a ship which is intended primarily to carry dry cargo in bulk, including such types as ore carriers as defined in regulation 1 of chapter XII of the International Convention for the Safety of Life at Sea (SOLAS), 1974, (as amended) but excluding combination carriers.

.6 Calendar year means the period from 1 January until 31 December inclusive.

the International Convention for the Safety of Life at Sea (SOLAS), 1974, (as amended) but excluding combination carriers.

.6 Calendar year means the period from 1 January until 31 December inclusive.

.7 Combination carrier means a ship designed to load 100% deadweight with both liquid and dry cargo in bulk.

.8 Company means the owner of the ship or any other organization or person such as the manager, or the bareboat charterer, who has assumed the responsibility for operation of the ship from the owner of the ship and who on assuming such responsibility has agreed to take over all the duties and responsibilities imposed by the International Management Code for the Safe Operation of Ships and for Pollution Prevention, as amended.

.9 Containership means a ship designed exclusively for the carriage of containers in holds and on deck.

.10 Conventional propulsion means a method of propulsion where a main

Operation of Ships and for Pollution Prevention, as amended.

.9 Containership means a ship designed exclusively for the carriage of containers in holds and on deck.

.10 Conventional propulsion means a method of propulsion where a main reciprocating internal combustion engine(s) is the prime mover and coupled to a propulsion shaft either directly or through a gear box.

.11 Cruise passenger ship means a passenger ship not having a cargo deck, designed exclusively for commercial transportation of passengers in overnight accommodations on a sea voyage.

.12 Distance travelled means distance travelled over ground.

.13 Existing ship means a ship which is not a new ship.

.14 Gas carrier means a cargo ship, other than an LNG carrier as defined in paragraph 2.16 of this regulation, constructed or adapted and used for the carriage in bulk of any liquefied gas.

.15 General cargo ship means a ship with a multi-deck or single deck hull designed primarily for the carriage of general cargo. This definition excludes specialized dry cargo ships, which are not included in the calculation of reference lines for

primarily for the carriage of general cargo. This definition excludes specialized dry cargo ships, which are not included in the calculation of reference lines for general cargo ships, namely livestock carrier, barge carrier, heavy load carrier, yacht carrier, nuclear fuel carrier.

.16 LNG carrier means a cargo ship constructed or adapted and used for the carriage in bulk of liquefied natural gas (LNG).

.17 Major conversion means a conversion of a ship:

.1 which substantially alters the dimensions, carrying capacity or engine power of the ship; or

.2 which changes the type of the ship; or

.3 the intent of which in the opinion of the Administration is substantially to prolong the life of the ship; or

.4 which otherwise so alters the ship that, if it were a new ship, it would become subject to relevant provisions of the present Convention not applicable to it as an existing ship; or

.5 which substantially alters the energy efficiency of the ship and includes any modifications that could cause the ship to exceed the applicable required EEDI as set out in regulation 24 of this Annex or

.5 which substantially alters the energy efficiency of the ship and includes any modifications that could cause the ship to exceed the applicable required EEDI as set out in regulation 24 of this Annex or the applicable required EEXI as set out in regulation 25 of this Annex.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 8 -

.18 New ship means a ship:

.1 for which the building contract is placed on or after 1 January 2013; or

.2 in the absence of a building contract, the keel of which is laid or which is at a similar stage of construction on or after 1 July 2013; or

.3 the delivery of which is on or after 1 July 2015.

.19 Non-conventional propulsion means a method of propulsion, other than conventional propulsion, including diesel-electric propulsion, turbine propulsion, and hybrid propulsion systems.

.20 Passenger ship means a ship which carries more than 12 passengers.

.21 Polar Code means the International Code for Ships Operating in Polar

and hybrid propulsion systems.

.20 Passenger ship means a ship which carries more than 12 passengers.

.21 Polar Code means the International Code for Ships Operating in Polar Waters, consisting of an introduction, parts I -A and II -A and parts I -B and II-B, as adopted by resolutions MSC.385(94) and MEPC.264(68), as may be amended, provided that:

.1 amendments to the environment -related provisions of the introduction and chapter 1 of part II-A of the Polar Code are adopted, brought into force and take effect in accordance with the provisions of article 16 of the present Convention concerning the amend ment procedures applicable to an appendix to an annex; and

.2 amendments to part II -B of the Polar Code are adopted by the Marine Environment Protection Committee in accordance with its Rules of Procedure.

.22 Refrigerated cargo carrier means a ship designed exclusively for the carriage of refrigerated cargoes in holds.

.23 Required annual operational CII is the target value of attained annual operational CII in accordance with regulations 26 and 28 of this Annex for the specific ship type and size.

of refrigerated cargoes in holds.

.23 Required annual operational CII is the target value of attained annual operational CII in accordance with regulations 26 and 28 of this Annex for the specific ship type and size.

.24 Required EEDI is the maximum value of attained EEDI that is allowed by regulation 24 of this Annex for the specific ship type and size.

.25 Required EEXI is the maximum value of attained EEXI that is allowed by regulation 25 of this Annex for the specific ship type and size.

.26 Ro-ro cargo ship means a ship designed for the carriage of roll -on-roll-off cargo transportation units.

.27 Ro-ro cargo ship (vehicle carrier) means a multi-deck roll-on-roll-off cargo ship designed for the carriage of empty cars and trucks.

.28 Ro-ro passenger ship means a passenger ship with roll -on-roll-off cargo spaces.

.29 Tanker means an oil tanker as defined in regulation 1 of Annex I of the present

.29 Tanker means an oil tanker as defined in regulation 1 of Annex I of the present Convention or a chemical tanker or an NLS tanker as defined in regulation 1 of Annex II to the present Convention.

Regulation 3
Exceptions and exemptions

General

1 Regulations of this Annex shall not apply to:

.1 any emission necessary for the purpose of securing the safety of a ship or saving life at sea; or

.2 any emission resulting from damage to a ship or its equipment:

.2.1 provided that all reasonable precautions have been taken after the occurrence of the damage or discovery of the emission for the purpose of preventing or minimizing the emission; and

.2.2 except if the owner or the master acted either with intent to cause damage, or recklessly and with knowledge that damage would probably result.

Trials for ship emission reduction and control technology research

2 The Administration of a Party may, in cooperation with other Administrations as

damage, or recklessly and with knowledge that damage would probably result.

Trials for ship emission reduction and control technology research

2 The Administration of a Party may, in cooperation with other Administrations as appropriate, issue an exemption from specific provisions of this Annex for a ship to conduct trials for the development of ship emission reduction and control technologies and engine design programmes. Such an exemption shall only be provided if the applications of specific provisions of the Annex or the revised NO x Technical Code 2008 could impede research into the development of such technologies or programmes. A permit issued under this regulation shall not exempt a ship from the reporting requirement under regulation 27 and shall not alter the type and scope of data required to be reported under regulation 27. A permit for such an exemption shall only be provided to the minimum number of ships necessary and be subject to the following provisions:

.1 for marine diesel engines with a per cylinder displacement up to 30 L, the duration of the sea trial shall not exceed 18 months. If additional time is

following provisions:

.1 for marine diesel engines with a per cylinder displacement up to 30 L, the duration of the sea trial shall not exceed 18 months. If additional time is required, a permitting Administration or Administrations may permit a renewal for one additional 18-month period; or

.2 for marine diesel engines with a per cylinder displacement at or above 30 L, the duration of the ship trial shall not exceed five years and shall require a progress review by the permitting Administration or Administrations at each intermediate survey. A permit may be withdrawn based on this review if the testing has not adhered to the conditions of the permit or if it is determined that the technology or programme is not likely to produce effective results in the reduction and control of ship emissions. If the reviewing Administration or Administrations determine that additional time is required to conduct a test of a particular technology or programme, a permit may be renewed for an additional time period not to exceed five years.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE

a particular technology or programme, a permit may be renewed for an additional time period not to exceed five years.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 10 -

Emissions from seabed mineral activities

3.1 Emissions directly arising from the exploration, exploitation and associated offshore processing of seabed mineral resources are, consistent with article 2(3)(b)(ii) of the present Convention, exempt from the provisions of this Annex. Such emissions include the following:

.1 emissions resulting from the incineration of substances that are solely and directly the result of exploration, exploitation and associated offshore processing of seabed mineral resources, including but not limited to the flaring of hydrocarbons and the burning of cuttings, muds, and/or stimulation fluids during well completion and testing operations, and flaring arising from upset conditions;

.2 the release of gases and volatile compounds entrained in drilling fluids and cuttings;

during well completion and testing operations, and flaring arising from upset conditions;

.2 the release of gases and volatile compounds entrained in drilling fluids and cuttings;

.3 emissions associated solely and directly with the treatment, handling or storage of seabed minerals; and

.4 emissions from marine diesel engines that are solely dedicated to the exploration, exploitation and associated offshore processing of seabed mineral resources.

3.2 The requirements of regulation 18 of this Annex shall not apply to the use of hydrocarbons that are produced and subsequently used on site as fuel, when approved by the Administration.

Unmanned non-self-propelled barges

4 The Administration may exempt an unmanned non-self-propelled (UNSP) barge from the requirements of regulations 5.1 and 6.1 of this Annex by means of an International Air Pollution Prevention Exemption Certificate for Unmanned Non-self-propelled (UNSP) Barges, for a period not exceeding five years provided that the barge has unde rgone a survey to confirm that conditions referred to in regulations 2.1.32.1 to 2.1.32.3 of this Annex are met.

Regulation 4
Equivalents

for a period not exceeding five years provided that the barge has unde rgone a survey to confirm that conditions referred to in regulations 2.1.32.1 to 2.1.32.3 of this Annex are met.

Regulation 4
Equivalents

1 The Administration of a Party may allow any fitting, material, appliance or apparatus to be fitted in a ship or other procedures, alternative fuel oils, or compliance methods used as an alternative to those required by this Annex if such fitting, material, appliance or apparatus or other procedures, alternative fuel oils, or compliance methods are at least as effective in terms of emissions reductions as those required by this Annex, including any of the standards set forth in regulations 13 and 14.

2 The Administration of a Party that allows a fitting, material, appliance or apparatus or other procedures, alternative fuel oils, or compliance methods used as an alternative to those required by this Annex shall communicate to the Organization for circulation to the Parties particulars thereof, for their information and appropriate action, if any.

3 The Administration of a Party should take into account any relevant guidelines

particulars thereof, for their information and appropriate action, if any.

3 The Administration of a Party should take into account any relevant guidelines developed by the Organization pertaining to the equivalents provided for in this regulation.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 11 -

4 The Administration of a Party that allows the use of an equivalent as set forth in paragraph 1 of this regulation shall endeavour not to impair or damage its environment, human health, property or resources or those of other States.

Chapter 2 Survey, certification and means of control

Regulation 5
Surveys

1 Every ship of 400 gross tonnage and above and every fixed and floating drilling rig or other platform shall, to ensure compliance with the requirements of chapter 3 of this Annex, be subject to the surveys specified below:

.1 An initial survey before the ship is put into service or before the certificate required under regulation 6 of this Annex is issued for the first time.

subject to the surveys specified below:

.1 An initial survey before the ship is put into service or before the certificate required under regulation 6 of this Annex is issued for the first time. This survey shall be such as to ensure that the equipment, systems, fittings, arrangements and material fully comply with the applicable requirements of chapter 3 of this Annex;

.2 A renewal survey at intervals specified by the Administration, but not exceeding five years, except where regulation 9.2, 9.5, 9.6 or 9.7 of this Annex is applicable. The renewal survey shall be such as to ensure that the equipment, systems, fittings, arrangements and material fully comply with applicable requirements of chapter 3 of this Annex;

.3 An intermediate survey within three months before or after the second anniversary date or within three months before or after the third anniversary date of the certificate which shall take the place of one of the annual surveys specified in paragraph 1.4 of this regulation. The intermediate survey shall be such as to ensure that the equipment and arrangements fully comply with the

date of the certificate which shall take the place of one of the annual surveys specified in paragraph 1.4 of this regulation. The intermediate survey shall be such as to ensure that the equipment and arrangements fully comply with the applicable requirements of chapter 3 of this Annex and are in good working order. Such intermediate surveys shall be endorsed on the IA PP Certificate issued under regulation 6 or 7 of this Annex;

.4 An annual survey within three months before or after each anniversary date of the certificate, including a general inspection of the equipment, systems, fittings, arrangements and material referred to in paragraph 1.1 of this regulation to ensure that they have been maintained in accordance with paragraph 5 of this regulation and that they remain satisfactory for the service for which the ship is intended. Such annual surveys shall be endorsed on the IAPP Certificate issued under regulation 6 or 7 of this Annex; and

.5 An additional survey either general or partial, according to the circumstances, shall be made whenever any important repairs or renewals are made as

IAPP Certificate issued under regulation 6 or 7 of this Annex; and

.5 An additional survey either general or partial, according to the circumstances, shall be made whenever any important repairs or renewals are made as prescribed in paragraph 5 of this regulation or after a repair resulting from investigations prescribed in paragraph 6 of this regulation. The survey shall be such as to ensure that the necessary repairs or renewals have been effectively made, that the material and workmanship of such repairs or renewals are in all respects satisfactory and that the ship complies in all respects with the requirements of chapter 3 of this Annex.

2 In the case of ships of less than 400 gross tonnage, the Administration may establish appropriate measures in order to ensure that the applicable provisions of chapter 3 of this Annex are complied with.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 12 -

3 Surveys of ships as regards the enforcement of the provisions of this Annex shall be

3 Surveys of ships as regards the enforcement of the provisions of this Annex shall be carried out by officers of the Administration.

.1 The Administration may, however, entrust the surveys either to surveyors nominated for the purpose or to organizations recognized by it.
Such organizations shall comply with the guidelines adopted by the Organization.

.2 The survey of marine diesel engines and equipment for compliance with regulation 13 of this Annex shall be conducted in accordance with the revised NOx Technical Code 2008.

.3 When a nominated surveyor or recognized organization determines that the condition of the equipment does not correspond substantially with the particulars of the certificate, it shall ensure that corrective action is taken and shall in due course notify the Administration. If such corrective action is not taken, the certificate shall be withdrawn by the Administration. If the ship is in a port of another Party, the appropriate authorities of the port State shall also be

taken, the certificate shall be withdrawn by the Administration. If the ship is in a port of another Party, the appropriate authorities of the port State shall also be notified immediately. When an officer of the Administration, a nominated surveyor or recognized organization has notified the appropriate authorities of the port State, the Government of the port State concerned shall give such officer, surveyor or organization any necessary assistance to carry out their obligations under this regulation.

.4 In every case, the Administration concerned shall fully guarantee the completeness and efficiency of the survey and shall undertake to ensure the necessary arrangements to satisfy this obligation.

4 Ships to which chapter 4 of this Annex applies shall also be subject to the surveys specified below, taking into account the guidelines adopted by the Organization:

.1 an initial survey carried out before a new ship is put in service and before the International Energy Efficiency Certificate is issued. The survey shall verify that the ship's attained EEDI is in accordance with the requirements in

.1 an initial survey carried out before a new ship is put in service and before the International Energy Efficiency Certificate is issued. The survey shall verify that the ship's attained EEDI is in accordance with the requirements in chapter 4 of this Annex, and that the SEEMP required by regulation 26 of this Annex is on board;

.2 a general or partial survey, according to the circumstances, carried out after a major conversion of a new ship to which this regulation applies. The survey shall ensure that the attained EEDI is recalculated as necessary and meets the requirement of regulation 24 of this Annex, with the reduction factor applicable to the ship type and size of the converted ship in the phase corresponding to the date of contract or keel laying or delivery determined for the original ship in accordance with regulation 2.2.18 of this Annex;

.3 in cases where the major conversion of a new or existing ship is so extensive that the ship is regarded by the Administration as a newly constructed ship, the Administration shall determine the necessity of an initial survey on attained

.3 in cases where the major conversion of a new or existing ship is so extensive that the ship is regarded by the Administration as a newly constructed ship, the Administration shall determine the necessity of an initial survey on attained EEDI. Such a survey, if determined necessary, shall ensure that the attained EEDI is calculated and meets the requirement of regulation 24 of this Annex, with the reduction factor applicable corresponding to the ship type and size of the converted ship at the date of the contract of the conversion, or in the absence of a contract, the commencement date of the conversion. The survey shall also verify that the SEEMP required by regulation 26 of this Annex is on board and, for a ship to which regulation 27 applies, has been revised

appropriately to reflect a major conversion in those cases where the major conversion affects data collection methodology and/or reporting processes;

appropriately to reflect a major conversion in those cases where the major conversion affects data collection methodology and/or reporting processes;

.4 for existing ships, the verification of the requirement to have a SEEMP on board according to regulation 26 of this Annex shall take place at the first intermediate or renewal survey identified in paragraph 1 of this regulation, whichever is the first, on or after 1 January 2013;

.5 the Administration shall ensure that for each ship to which regulation 27 applies, the SEEMP complies with regulation 26.2 of this Annex. This shall be done prior to collecting data under regulation 27 of this Annex in order to ensure the methodology and processes are in place prior to the beginning of the ship's first reporting period. Confirmation of compliance shall be provided to and retained on board the ship;

.6 the Administration shall ensure that, for each ship to which regulation 28 applies, the SEEMP complies with regulation 26.3.1 of this Annex. This shall be done prior to 1 January 2023. Confirmation of compliance shall be

.6 the Administration shall ensure that, for each ship to which regulation 28 applies, the SEEMP complies with regulation 26.3.1 of this Annex. This shall be done prior to 1 January 2023. Confirmation of compliance shall be provided to, and retained on board, the ship;

.7 the verification that the ship 's attained EEXI is in accordance with the requirements in regulations 23 and 25 of this Annex shall take place at the first annual, intermediate or renewal survey identified in paragraph 1 of this regulation or the initial survey identified in paragraphs 4.1 and 4.3 of this regulation, whichever is the first, on or after 1 January 2023; and

.8 notwithstanding paragraph 4.7 of this regulation, a general or partial survey, according to the circumstances, carried out after a major conversion of a ship to which regulation 23 of this Annex applies. The survey shall ensure that the attained EEXI is recalculated as necessary and meets the requirement of regulation 25 of this Annex.

5 The equipment shall be maintained to conform with the provisions of this Annex and no changes shall be made in the equipment, systems, fittings, arrangements or material covered by

regulation 25 of this Annex.

5 The equipment shall be maintained to conform with the provisions of this Annex and no changes shall be made in the equipment, systems, fittings, arrangements or material covered by the survey, without the express approval of the Administration. The direct replacement of such equipment and fittings with equipment and fittings that conform with the provisions of this Annex is permitted.

6 Whenever an accident occurs to a ship or a defect is discovered that substantially affects the efficiency or completeness of its equipment covered by this Annex, the master or owner of the ship shall report at the earliest opportunity to the Administration, a nominated surveyor or recognized organization responsible for issuing the relevant certificate.

Regulation 6
Issue or endorsement of Certificates and Statements of Compliance related to fuel oil consumption reporting and operational carbon intensity rating

International Air Pollution Prevention Certificate

1 An International Air Pollution Prevention (IAPP) Certificate shall be issued, after an initial or renewal survey in accordance with the provisions of regulation 5 of this Annex, to:

International Air Pollution Prevention Certificate

1 An International Air Pollution Prevention (IAPP) Certificate shall be issued, after an initial or renewal survey in accordance with the provisions of regulation 5 of this Annex, to:

.1 any ship of 400 gross tonnage and above engaged in voyages to ports or offshore terminals under the jurisdiction of other Parties; and

.2 platforms and drilling rigs engaged in voyages to waters under the sovereignty or jurisdiction of other Parties.

2 A ship constructed before the date this Annex enters into force for that particular ship's Administration, shall be issued with an IAPP Certificate in accordance with paragraph 1 of this regulation no later than the first scheduled dry-docking after the date of such entry into force, but in no case later than three years after this date.

3 Such certificate shall be issued or endorsed either by the Administration or by any person

but in no case later than three years after this date.

3 Such certificate shall be issued or endorsed either by the Administration or by any person or organization duly authorized by it. In every case, the Administration assumes full responsibility for the certificate.

International Energy Efficiency Certificate

4 An International Energy Efficiency Certificate for the ship shall be issued after a survey in accordance with the provisions of regulation 5.4 of this Annex to any ship of 400 gross tonnage and above before that ship may engage in voyages to ports or offshore terminals under the jurisdiction of other Parties.

5 The certificate shall be issued or endorsed either by the Administration or any organization duly authorized by it. In every case, the Administration assumes full responsibility for the certificate.

Statement of Compliance related to fuel oil consumption reporting and operational carbon intensity rating

6 Upon receipt of reported data pursuant to regulation 27.3 of this Annex and attained annual operational CII pursuant to regulation 28.2 of this Annex, the Administration or any organization duly authorized by it shall:

6 Upon receipt of reported data pursuant to regulation 27.3 of this Annex and attained annual operational CII pursuant to regulation 28.2 of this Annex, the Administration or any organization duly authorized by it shall:

.1 determine whether the data has b een reported in accordance with regulation 27 of this Annex;

.2 verify that the attained annual operational CII reported is based on the data submitted in accordance with regulation 27 of this Annex;

.3 based on the verified attained annual operation al CII, determine the operational carbon intensity rating of the ship in accordance with regulation 28.6 of this Annex; and

.4 issue a Statement of Compliance related to fuel oil consumption reporting and operational carbon intensity rating to the ship no later than five months from the beginning of the calendar year, upon determination and verification pursuant to regulations 6.6.1 to 6.6.3 of this Annex . In every case, the Administration assumes full responsibility for this Statement of Compliance.

7 Upon receipt of reported data pursuant to regulations 27.4, 27.5 or 27.6 of this Annex,

pursuant to regulations 6.6.1 to 6.6.3 of this Annex . In every case, the Administration assumes full responsibility for this Statement of Compliance.

7 Upon receipt of reported data pursuant to regulations 27.4, 27.5 or 27.6 of this Annex, the Administration or any organization duly authorized by it shall promptly determine whether the data has been reported in accordance with regulation 27 and, if so, is sue a Statement of Compliance to the ship. In every case, the Administration assumes full responsibility for this Statement of Compliance.

8 Notwithstanding paragraph 6 of this regulation, a ship rated as D for three consecutive years or rated as E in acc ordance with regulation 28 of this Annex shall not be issued a Statement of Compliance unless a plan of corrective actions is duly developed and reflected

in the SEEMP and verified by the Administration or any organization duly authorized by it in

in the SEEMP and verified by the Administration or any organization duly authorized by it in accordance with regulations 28.7 and 28.8 of this Annex.

Regulation 7
Issue of a Certificate by another Party

1 A Party may, at the request of the Administration, cause a ship to be surveyed and, if satisfied that the provisions of this Annex are complied with, shall issue or authorize the issue of an IAPP Certificate or an International Energy Efficiency Certificate to the ship, and where appropriate, endorse or authorize the endorsement of such certificates on the ship, in accordance with this Annex.

2 A copy of the certificate and a copy of the survey report shall be transmitted as soon as possible to the requesting Administration.

3 A certificate so issued shall contain a statement to the effect that it has been issued at the request of the Administration and it shall have the same force and receive the same recognition as a certificate issued under regulation 6 of this Annex.

at the request of the Administration and it shall have the same force and receive the same recognition as a certificate issued under regulation 6 of this Annex.

4 No IAPP Certificate, International Energy Efficiency Certificate or UNSP Exemption Certificate shall be issued to a ship which is entitled to fly the flag of a State which is not a Party.

Regulation 8
Form of Certificates and Statements of Compliance related to fuel oil consumption reporting and operational carbon intensity rating

International Air Pollution Prevention Certificate

1 The IAPP Certificate shall be drawn up in a form corresponding to the model given in appendix I to this Annex and shall be at least in English, French or Spanish. If an official language of the issuing country is also used, this shall prevail in case of a dispute or discrepancy.

International Energy Efficiency Certificate

2 The International Energy Efficiency Certificate shall be drawn up in a form corresponding to the model given in appendix VIII to this Annex and shall be at least in English, French or Spanish. If an official language of the issuing Party is also used, this shall prevail in

corresponding to the model given in appendix VIII to this Annex and shall be at least in English, French or Spanish. If an official language of the issuing Party is also used, this shall prevail in case of a dispute or discrepancy.

Statement of Compliance related to fuel oil consumption reporting and operational carbon intensity rating

3 The Statement of Compliance pursuant to regulations 6.6 and 6.7 of this Annex shall be drawn up in a form corresponding to the model given in appendix X to this Annex and shall be at least in English, French or Spanish. If an official language of the issuing Party is also used, this shall prevail in case of a dispute or discrepancy.

International Air Pollution Prevention Exemption Certificate for Unmanned Non-self-propelled Barges

4 In accordance with regulation 3.4 of this Annex, the International Air Pollution Prevention Exemption Certificate for Unmanned Non-self-propelled Barges shall be drawn up in the form corresponding to the model given in appendix XI to this Annex and shall be at least in English, French or Spanish. If an official language of the issuing country is also used, this shall prevail in the

corresponding to the model given in appendix XI to this Annex and shall be at least in English, French or Spanish. If an official language of the issuing country is also used, this shall prevail in the event of a dispute or discrepancy.
RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 16 -

Regulation 9
Duration and validity of Certificates and Statements of Compliance related to fuel oil consumption reporting and operational carbon intensity rating

International Air Pollution Prevention Certificate

1 An IAPP Certificate shall be issued for a period specified by the Administration, which shall not exceed five years.

2 Notwithstanding the requirements of paragraph 1 of this regulation:

.1 when the renewal survey is completed within three months before the expiry date of the existing certificate, the new certificate shall be valid from the date of completion of the renewal survey to a date not exceeding five years from the date of expiry of the existing certificate;

date of the existing certificate, the new certificate shall be valid from the date of completion of the renewal survey to a date not exceeding five years from the date of expiry of the existing certificate;

.2 when the renewal survey is completed after the expiry date of the existing certificate, the new certificate shall be valid from the date of completion of the renewal survey to a date not exceeding five years from the date of expiry of the existing certificate; and

.3 when the renewal survey is completed more than three months before the expiry date of the existing certificate, the new certificate shall be valid from the date of completion of the renewal survey to a date not exceeding five years from the date of completion of the renewal survey.

3 If a certificate is issued for a period of less than five years, the Administration may extend the validity of the certificate beyond the expiry date to the maximum period specified in paragraph 1 of this regulation, provided that the surveys referred to in regulations 5.1.3 and 5.1.4 of this Annex applicable when a certificate is issued for a period of five years are carried out as appropriate.

paragraph 1 of this regulation, provided that the surveys referred to in regulations 5.1.3 and 5.1.4 of this Annex applicable when a certificate is issued for a period of five years are carried out as appropriate.

4 If a renewal survey has been completed and a new certificate cannot be issued or placed on board the ship before the expiry date of the existing certificate, the person or organization authorized by the Administration may endorse the existing certificate and such a certificate shall be accepted as valid for a further period that shall not exceed five months from the expiry date.

5 If a ship, at the time when a certificate expires, is not in a port in which it is to be surveyed, the Administration may extend the period of validity of the certificate, but this extension shall be granted only for the purpose of allowing the ship to complete its voyage to the port in which it is to be surveyed, and then only in cases where it appears proper and reasonable to do so. No certificate shall be extended for a period longer than three months, and a ship to which an extension is granted shall not, on its arrival in the port in which it is to

reasonable to do so. No certificate shall be extended for a period longer than three months, and a ship to which an extension is granted shall not, on its arrival in the port in which it is to be surveyed, be entitled by virtue of such extension to leave that port without having a new certificate. When the renewal survey is completed, the new certificate shall be valid to a date not exceeding five years from the date of expiry of the existing certificate before the extension was granted.

6 A certificate issued to a ship engaged on short voyages that has not been extended under the foregoing provisions of this regulation may be extended by the Administration for a period of grace of up to one month from the date of expiry stated on it. When the renewal survey is completed, the new certificate shall be valid to a date not exceeding five years from the date of expiry of the existing certificate before the extension was granted.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 17 -

7 In special circumstances, as determined by the Administration, a new certificate need not be dated from the date of expiry of the existing certificate as required by paragraph 2.1, 5 or 6 of this regulation. In these special circumstances, the new certificate shall be valid to a date not exceeding five years from the date of completion of the renewal survey.

8 If an annual or intermediate survey is completed before the period specified in regulation 5 of this Annex, then:

.1 the anniversary date shown on the certificate shall be amended by endorsement to a date that shall not be more than three months later than the date on which the survey was completed;

.2 the subsequent annual or intermediate survey required by regulation 5 of this Annex shall be completed at the intervals prescribed by that regulation using the new anniversary date; and

.3 the expiry date may remain unchanged, provided one or more annual or

Annex shall be completed at the intervals prescribed by that regulation using the new anniversary date; and

.3 the expiry date may remain unchanged, provided one or more annual or intermediate surveys, as appropriate, are carried out so that the maximum intervals between the surveys prescribed by regulation 5 of this Annex are not exceeded.

9 A certificate issued under regulation 6 or 7 of this Annex shall cease to be valid in any of the following cases:

.1 if the relevant surveys are not completed within the periods specified under regulation 5.1 of this Annex;

.2 if the certificate is not endorsed in accordance with regulation 5.1.3 or 5.1.4 of this Annex; and

.3 upon transfer of the ship to the flag of another State. A new certificate shall only be issued when the Government issuing the new certificate is fully satisfied that the ship is in compliance with the requirements of regulation 5.4 of this Annex. In the case of a transfer between Parties, if requested within three months after the transfer has taken place, the Government of the Party whose flag the ship was formerly entitled to fly shall, as soon as possible,

of this Annex. In the case of a transfer between Parties, if requested within three months after the transfer has taken place, the Government of the Party whose flag the ship was formerly entitled to fly shall, as soon as possible, transmit to the Administration copies of the certificate carried by the ship before the transfer and, if available, copies of the relevant survey reports.

International Energy Efficiency Certificate

10 The International Energy Efficiency Certificate shall be valid throughout the life of the ship subject to the provisions of paragraph 11 below.

11 An International Energy Efficiency Certificate issued under this Annex shall cease to be valid in any of the following cases:

.1 if the ship is withdrawn from service or if a new certificate is issued following major conversion of the ship; or

.2 upon transfer of the ship to the flag of another State. A new certificate shall only be issued when the Government issuing the new certificate is fully satisfied that the ship is in compliance with the requirements of chapter 4 of this Annex. In the case of a transfer between Parties, if requested within three

only be issued when the Government issuing the new certificate is fully satisfied that the ship is in compliance with the requirements of chapter 4 of this Annex. In the case of a transfer between Parties, if requested within three months after the transfer has taken place, the Government of the Party

whose flag the ship was formerly entitled to fly shall, as soon as possible, transmit to the Administration copies of the certificate carried by the ship before the transfer and, if available, copies of the relevant survey reports; or

.3 if the ship's equipment, systems, fittings, arrangements, or material covered by the survey were changed without the express approval of the Administration, as provided for in regulation 5.5 of this Annex, unless regulation 3 of this Annex applies.

Statement of Compliance related to fuel oil consumption reporting and operational carbon intensity rating

Administration, as provided for in regulation 5.5 of this Annex, unless regulation 3 of this Annex applies.

Statement of Compliance related to fuel oil consumption reporting and operational carbon intensity rating

12 The Statement of Compliance issued pursuant to regulation 6.6 of this Annex shall be valid for the calendar year in which it is issued and for the first five months of the following calendar year. The Statement of Compliance issued pursuant to regulation 6.7 of this Annex shall be valid for the calendar year in which it is issued, for the following calendar year, and for the first five months of the subsequent calendar year. All Statements of Compliance shall be kept on board for at least five years.

Regulation 10
Port State control on operational requirements

1 A ship, when in a port or an offshore terminal under the jurisdiction of another Party, is subject to inspection by officers duly authorized by such Party concerning operational requirements under this Annex, where there are clear grounds for believing that the master or crew are not familiar with essential shipboard procedures relating to the prevention of air pollution from ships.

requirements under this Annex, where there are clear grounds for believing that the master or crew are not familiar with essential shipboard procedures relating to the prevention of air pollution from ships.

2 In the circumstances given in paragraph 1 of this regulation, the Party shall take steps to ensure that the ship shall not sail until the situation has been brought to order in accordance with the requirements of this Annex.

3 Procedures relating to the port State control prescribed in article 5 of the present Convention shall apply to this regulation.

4 Nothing in this regulation shall be construed to limit the rights and obligations of a Party carrying out control over operational requirements specifically provided for in the present Convention.

5 In relation to chapter 4 of this An nex, any port State inspection may verify, when appropriate, that there is a valid Statement of Compliance related to fuel oil consumption reporting and operational carbon intensity rating, an International Energy Efficiency Certificate and a Ship Energy Efficiency Management Plan on board, in accordance with article 5 of the present Convention.

reporting and operational carbon intensity rating, an International Energy Efficiency Certificate and a Ship Energy Efficiency Management Plan on board, in accordance with article 5 of the present Convention.

6 Notwithstanding the requirements in paragraph 5 of this regulation, any port State inspection may inspect whether the Ship Energy Efficiency Management Plan is duly implemented by the ship in accordance with regulation 28 of this Annex.

Regulation 11
Detection of violations and enforcement

1 Parties shall cooperate in the detection of violations and the enforcement of the provisions of this Annex, using all appropriate and practicable measures of detection and environmental monitoring, and adequate procedures for reporting and accumulation of evidence.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 19 -

2 A ship to which this Annex applies may, in any port or offshore terminal of a Party, be subject to inspection by officers appointed or authorized by that Party for the purpose of

2 A ship to which this Annex applies may, in any port or offshore terminal of a Party, be subject to inspection by officers appointed or authorized by that Party for the purpose of verifying whether the ship has emitted any of the substances covered by this Annex in violation of the provision of this Annex. If an inspection indicates a violation of this Annex, a report shall be forwarded to the Administration for any appropriate action.

3 Any Party shall furnish to the Administration evidence, if any, that the ship has emitted any of the substances covered by this Annex in violation of the provisions of this Annex. If it is practicable to do so, the competent authority of the former Party shall notify the master of the ship of the alleged violation.

4 Upon receiving such evidence, the Administration shall investigate the matter and may request the other Party to furnish further or better evidence of the alleged contravention. If the Administration is satisfied that sufficient evidence is available to enable proceedings to be brought in respect of the alleged violation, it shall cause such proceedings to be taken in

If the Administration is satisfied that sufficient evidence is available to enable proceedings to be brought in respect of the alleged violation, it shall cause such proceedings to be taken in accordance with its law as soon as possible. The Administration shall promptly inform the Party that has reported the alleged violation, as well as the Organization, of the action taken.

5 A Party may also inspect a ship to which this Annex appl ies when it enters the ports or offshore terminals under its jurisdiction, if a request for an investigation is received from any Party together with sufficient evidence that the ship has emitted any of the substances covered by the Annex in any place in violation of this Annex. The report of such investigation shall be sent to the Party requesting it and to the Administration so that the appropriate action may be taken under the present Convention.

6 The international law concerning the prevention, reduction and control of pollution of the marine environment from ships, including that law relating to enforcement and safeguards, in force at the time of application or interpretation of this Annex, applies, mutatis mutandis, to

the marine environment from ships, including that law relating to enforcement and safeguards, in force at the time of application or interpretation of this Annex, applies, mutatis mutandis, to the rules and standards set forth in this Annex.

Chapter 3 Requirements for control of emissions from ships

Regulation 12
Ozone-depleting substances

1 This regulation does not apply to permanently sealed equipment where there are no refrigerant charging connections or potentially removable components containing ozone-depleting substances.

2 Subject to the provisions of regulation 3.1, any deliberate emissions of ozone-depleting substances shall be prohibited. Deliberate emissions include emissions occurring in the course of maintaining, servicing, repairing or disposing of systems or equipment, except that deliberate emissions do not include minimal releases associated with the recapture or recycling of an ozone-depleting substance. Emissions arising from leaks of an ozone-depleting substance, whether or not the leaks are deliberate, may be regulated by Parties.

3.1 Installations that contain ozone -depleting substances, other than

an ozone-depleting substance, whether or not the leaks are deliberate, may be regulated by Parties.

3.1 Installations that contain ozone -depleting substances, other than hydrochlorofluorocarbons, shall be prohibited:

.1 on ships constructed on or after 19 May 2005; or

.2 in the case of ships constructed before 19 May 2005 which have a contractual delivery date of the equipment to the ship on or after 19 May 2005 or, in the absence of a contractual delivery date, the actual delivery of the equipment to the ship on or after 19 May 2005.

3.2 Installations that contain hydrochlorofluorocarbons shall be prohibited:

.1 on ships constructed on or after 1 January 2020; or

.2 in the case of ships constructed before 1 January 2020 which have a contractual delivery date of the equipment to the ship on or after 1 January 2020 or, in the absence of a contractual delivery date, the

.2 in the case of ships constructed before 1 January 2020 which have a contractual delivery date of the equipment to the ship on or after 1 January 2020 or, in the absence of a contractual delivery date, the actual delivery of the equipment to the ship on or after 1 January 2020.

4 The substances referred to in this regulation, and equipment containing such substances, shall be delivered to appropriate reception facilities when removed from ships.

5 Each ship subject to regulation 6.1 shall maintain a list of equipment containin g ozone-depleting substances.

6 Each ship subject to regulation 6.1 that has rechargeable systems that contain ozone-depleting substances shall maintain an ozone -depleting substances record book. This record book may form part of an existing logbook or electronic record book as approved by the Administration. An electronic recording system referred to in regulation 12.6, as adopted by resolution MEPC.176(58), shall be considered an electronic record book, provided the electronic recording syst em is approved by the Administration on or before the first IAPP

by resolution MEPC.176(58), shall be considered an electronic record book, provided the electronic recording syst em is approved by the Administration on or before the first IAPP Certificate renewal survey carried out on or after 1 October 2020, but not later than 1 October 2025, taking into account the guidelines developed by the Organization.

7 Entries in the ozone-depleting substances record book shall be recorded in terms of mass (kg) of substance and shall be completed without delay on each occasion, in respect of the following:

.1 recharge, full or partial, of equipment containing ozone -depleting substances;

.2 repair or maintenance of equipment containing ozone-depleting substances;

.3 discharge of ozone-depleting substances to the atmosphere:

.3.1 deliberate; and

.3.2 non-deliberate;

.4 discharge of ozone-depleting substances to land-based reception facilities; and

.5 supply of ozone-depleting substances to the ship.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 21 -

Regulation 13
Nitrogen oxides (NOx)

Application

1.1 This regulation shall apply to:

.1 each marine diesel engine with a power output of more than 130 kW installed on a ship; and

.2 each marine diesel engine with a power output of more than 130 kW that undergoes a major conversion on or after 1 January 2000 except when demonstrated to the satisfaction of the Administration that such engine is an identical replacement to the engine that it is replacing and is otherwise not covered under paragraph 1.1.1 of this regulation.

1.2 This regulation does not apply to:

.1 a marine diesel engine intended to be used solely for emergencies or solely to power any device or equipment intended to be used solely for emergencies on the ship on which it is installed, or a marine diesel engine installed in lifeboats intended to be used solely for emergencies; and

.2 a marine diesel engine installed on a ship solely engaged in voyages within

emergencies on the ship on which it is installed, or a marine diesel engine installed in lifeboats intended to be used solely for emergencies; and

.2 a marine diesel engine installed on a ship solely engaged in voyages within waters subject to the sovereignty or jurisdiction of the State the flag of which the ship is entitled to fly, provided that such engine is subject to an alternative NOx control measure established by the Administration.

1.3 Notwithstanding the provisions of paragraph 1.1 of this regulation, the Administration may provide an exclusion from the application of this regulation for any marine diesel engine that is installed on a ship constructed, or for any marine diesel engine that undergoes a major conversion, before 19 May 2005, provided that the ship on which the engine is installed is solely engaged in voyages to ports or offshore terminals within the State the flag of which the ship is entitled to fly.

Major conversion

2.1 For the purpose of this regulation, major conversion means a modification on or after 1 January 2000 of a marine diesel engine that has not already been certified to the

ship is entitled to fly.

Major conversion

2.1 For the purpose of this regulation, major conversion means a modification on or after 1 January 2000 of a marine diesel engine that has not already been certified to the standards set forth in paragraph 3, 4 or 5.1.1 of this regulation where:

.1 the engine is replaced by a marine diesel engine or an additional marine diesel engine is installed, or

.2 any substantial modification, as defined in the revised NOx Technical Code 2008, is made to the engine, or

.3 the maximum continuous rating of the engine is increased by more than 10% compared to the maximum continuous rating of the original certification of the engine.

2.2 For a major conversion involving the replacement of a marine diesel engine with a non-identical marine diesel engine, or the installation of an additional marine diesel engine, the standards in this regulation at the time of the replacement or addition of the engine shall apply. In the case of replacement engines only, if it is not possible for such a replacement engine to

In the case of replacement engines only, if it is not possible for such a replacement engine to

meet the standards set forth in paragraph 5.1.1 of this regulation (Tier III, as applicable), then that replacement engine shall meet the standards set forth in paragraph 4 of this regulation (Tier II), taking into account the guidelines developed by the Organization.

2.3 A marine diesel engine referred to in paragraph 2.1.2 or 2.1.3 of this regulation shall meet the following standards:

.1 for ships constructed prior to 1 January 2000, the standards set forth in paragraph 3 of this regulation shall apply; and

.2 for ships constructed on or after 1 January 2000, the standards in force at the time the ship was constructed shall apply.

Tier I

3 Subject to regulation 3 of this Annex, the operation of a marine diesel engine that is installed on a ship constructed on or after 1 January 2000 and prior to 1 January 2011 is

the time the ship was constructed shall apply.

Tier I

3 Subject to regulation 3 of this Annex, the operation of a marine diesel engine that is installed on a ship constructed on or after 1 January 2000 and prior to 1 January 2011 is prohibited, except when the emission of nitrogen oxides (calculated as the total weighted emission of NO2) from the engine is within the following limits, where n = rated engine speed (crankshaft revolutions per minute):

.1 17.0 g/kWh when n is less than 130 rpm;

.2 45 · n(0.2) g/kWh when n is 130 or more but less than 2,000 rpm; and

.3 9.8 g/kWh when n is 2,000 rpm or more.

Tier II

4 Subject to regulation 3 of this Annex, the operation of a marine diesel engine that is installed on a ship constructed on or after 1 January 2011 is prohibited, except when the emission of nitrogen oxides (calculated as the total weighted emission of NO2) from the engine is within the following limits, where n = rated engine speed (crankshaft revolutions per minute):

.1 14.4 g/kWh when n is less than 130 rpm;

.2 44 · n(0.23) g/kWh when n is 130 or more but less than 2,000 rpm;

.3 7.7 g/kWh when n is 2,000 rpm or more.

Tier III

.1 14.4 g/kWh when n is less than 130 rpm;

.2 44 · n(0.23) g/kWh when n is 130 or more but less than 2,000 rpm;

.3 7.7 g/kWh when n is 2,000 rpm or more.

Tier III

5.1 Subject to regulation 3 of this Annex, in an emission control area designated for Tier III NOx control under paragraph 6 of this regulation (NOx Tier III emission control area) , the operation of a marine diesel engine that is installed on a ship is prohibited:

.1 except when the emission of nitrogen oxides (calculated as the total weighted emission of NO2) from the engine is within the following limits, where n = rated engine speed (crankshaft revolutions per minute):

.1 3.4 g/kWh when n is less than 130 rpm;

.2 9 · n(0.2) g/kWh when n is 130 or more but less than 2,000 rpm;
.3 2.0 g/kWh when n is 2,000 rpm or more;
RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 23 -

when

.2 that ship is constructed on or after:

.1 1 January 2016 and is operating in the North American Emission

when

.2 that ship is constructed on or after:

.1 1 January 2016 and is operating in the North American Emission Control Area or the United States Caribbean Sea Emission Control Area;

.2 1 January 2021 and is operating in the Baltic Sea Emission Control Area or the North Sea Emission Control Area;

.3 that ship is operating in a NOx Tier III emission control area other than an emission control area described in paragraph 5.1.2 of this regulation, and is constructed on or after the date of adoption of such an emission control area, or a later date as may be specified in the amendment designating the NOx Tier III emission control area, whichever is later.

5.2 The standards set forth in paragraph 5.1.1 of this regulation shall not apply to:

.1 a marine diesel engine installed on a ship with a length (L), as defined in regulation 1.19 of Annex I to the present Convention, of less than 24 metres when it has been specifically designed, and is used solely, for recreational purposes; or

.2 a marine diesel engine installed on a ship with a combined nameplate diesel

when it has been specifically designed, and is used solely, for recreational purposes; or

.2 a marine diesel engine installed on a ship with a combined nameplate diesel engine propulsion power of less than 750 kW if it is demonstrated, to the satisfaction of the Administration, that the ship cannot comply with the standards set forth in paragraph 5.1.1 of this regulation because of design or construction limitations of the ship; or

.3 a marine diesel engine installed on a ship constructed prior to 1 January 2021 of less than 500 gross tonnage, with a length (L), as defined in regulation 1.19 of Annex I to the present Convention, of 24 metres or over when it has been specifically designed, and is used solely, for recreational purposes.

5.3 The tier and on/off status of marine diesel engines installed on board a ship to which paragraph 5.1 of this regulation applies which are certified to both Tier II and Tier III or which are certified to Tier II only shall be recorded in such logbook or electronic record book as prescribed by the Administration at entry into and exit from a NO x Tier III emission control

are certified to Tier II only shall be recorded in such logbook or electronic record book as prescribed by the Administration at entry into and exit from a NO x Tier III emission control area, or when the on/off status changes within such an area, together with the date, time and position of the ship.

5.4 Emissions of nitrogen oxides from a marine diesel engine subject to paragraph 5.1 of this regulation that occur immediately following building and sea trials of a newly constructed ship, or before and following converting, repairing, and/or maintaining the ship, or maintenance or repair of a Tier II engine or a dual fuel engine when the ship is required to not have gas fuel or gas cargo on board due to safety requirements, for which activities take place in a shipyard or other repair facility located in a NOx Tier III emission control area, are temporarily exempted provided the following conditions are met:

.1 the engine meets the Tier II NOx limits; and
.2 the ship sails directly to or from the shipyard or other repair facility, does not load or unload cargo during the duration of the exemption, and follows any

.1 the engine meets the Tier II NOx limits; and
.2 the ship sails directly to or from the shipyard or other repair facility, does not load or unload cargo during the duration of the exemption, and follows any

additional specific routeing requirements indicated by the port State in which the shipyard or other repair facility is located, if applicable.

5.5 The exemption described in paragraph 5.4 of this regulation applies only for the following period:

.1 for a newly constructed ship, the period beginning at the time the ship is delivered from the shipyard, including sea trials, and ending at the time the ship directly exits the NOx Tier III emission control area(s) or, with regard to a ship fitted with a d ual fuel engine, the ship directly exits the NO x Tier III emission control area(s) or proceeds directly to the nearest gas fuel bunkering facility appropriate to the ship located in the NOx Tier III emission control area(s);

emission control area(s) or proceeds directly to the nearest gas fuel bunkering facility appropriate to the ship located in the NOx Tier III emission control area(s);

.2 for a ship with a Tier II engine undergoing conversion, maintenance or repair, the period beginning at the time the ship enters the NO x Tier III emission control area(s) and proceeds directly to the shipyard or other repair facility, and ending at the time the ship is released from the shipyard or other repair facility and directly exits the NO x Tier III emission control area (s) after performing sea trials, if applicable; or

.3 for a ship with a dual fuel engine undergoing conversion, maintenance or repair, when the ship is required to not have gas fuel or gas cargo on board due to safety requirements, the period beginning at the time the ship enters the NOx Tier III emission control area(s) or when it is degassed in the NO x Tier III emission control area(s) and proceeds directly to the shipyard or other repair facility, and ending at the time when the ship is released from the shipyard or other repair facility and directly exits the NO x Tier III emission

Tier III emission control area(s) and proceeds directly to the shipyard or other repair facility, and ending at the time when the ship is released from the shipyard or other repair facility and directly exits the NO x Tier III emission control area(s) or proceeds directly to the nearest gas fuel bunkering facility appropriate to the ship located in the NOx Tier III emission control area(s).

Emission control area

6 For the purposes of this regulation, a NOx Tier III emission control area shall be any sea area, including any port area, designated by the Organization in accordance with the criteria and procedures set forth in appendix III to this Annex. The NOx Tier III emission control areas are:

.1 the North American Emission Control Area, which means the area described by the coordinates provided in appendix VII to this Annex;

.2 the United States Caribbean Sea Emission Control Area, which means the area described by the coordinates provided in appendix VII to this Annex;

.3 the Baltic Sea area as defined in regulation 1.11.2 of Annex I of the present Convention; and

.4 the North Sea area as defined in regulation 1.14.6 of Annex V of the present

.3 the Baltic Sea area as defined in regulation 1.11.2 of Annex I of the present Convention; and

.4 the North Sea area as defined in regulation 1.14.6 of Annex V of the present Convention.

Marine diesel engines installed on a ship constructed prior to 1 January 2000

7.1 Notwithstanding paragraph 1.1.1 of this regulation, a marine diesel engine with a power output of more than 5,000 kW and a per cylinder displacement at or above 90 L installed

on a ship constructed on or after 1 January 1990 but prior to 1 January 2000 shall comply with the emission limits set forth in paragraph 7.4 of this regulation, provided that an approved method for that engine has been certified by an Administration of a Party and notification of such certification has been submitted to the Organization by the certifying Administration. Compliance with this paragraph shall be demonstrated through one of the following:

such certification has been submitted to the Organization by the certifying Administration. Compliance with this paragraph shall be demonstrated through one of the following:

.1 installation of the certified approved method, as confirmed by a survey using the verification procedure specified in the approved method file, including appropriate notation on the ship's IAPP Certificate of the presence of the approved method; or

.2 certification of the engine confirming that it operates within the limits set forth in paragraph 3, 4, or 5.1.1 of this regulation and an appropriate notation of the engine certification on the ship's IAPP Certificate.

7.2 Paragraph 7.1 of this regulation shall apply no later than the first renewal survey that occurs 12 months or more after deposit of the notification in paragraph 7.1. If a shipowner of a ship on which an approved method is to be installed can demonstrate to the satisfaction of the Administration that the approved method was not commercially available despite best efforts to obtain it, then that approved method shall be installed on the ship no later than the

the Administration that the approved method was not commercially available despite best efforts to obtain it, then that approved method shall be installed on the ship no later than the next annual survey of that ship that falls after the approved method is commercially available.

7.3 With regard to a marine diesel engine with a power output of more than 5,000 kW and a per cylinder displacement at or above 90 L installed on a ship constructed on or after 1 January 1990, but prior to 1 January 2000, the IAPP Certificate shall, for a marine diesel engine to which paragraph 7.1 of this regulation applies, indicate one of the following:

.1 an approved method has been applied pursuant to paragraph 7.1.1 of this regulation;

.2 the engine has been certified pursuant to paragraph 7.1.2 of this regulation;

.3 an approved method is not yet commercially available as described in paragraph 7.2 of this regulation; or

.4 an approved method is not applicable.

7.4 Subject to regulation 3 of this Annex, the operation of a marine diesel engine described in paragraph 7.1 of this regulation is prohibited, except when the emission of nitrogen oxides

.4 an approved method is not applicable.

7.4 Subject to regulation 3 of this Annex, the operation of a marine diesel engine described in paragraph 7.1 of this regulation is prohibited, except when the emission of nitrogen oxides (calculated as the total weighted emission of NO2) from the engine is within the following limits, where n = rated engine speed (crankshaft revolutions per minute):

.1 17.0 g/kWh when n is less than 130 rpm;

.2 45 · n(0.2) g/kWh when n is 130 or more but less than 2,000 rpm; and

.3 9.8 g/kWh when n is 2,000 rpm or more.

7.5 Certification of an approved method shall be in accordance with chapter 7 of the revised NOx Technical Code 2008 and shall include verification:

.1 by the designer of the base marine diesel engine to which the approved method applies that the calculated effect of the approved method will not decrease engine rating by more than 1.0%, increase fuel consumption by more than 2.0% as measured according to the appropriate test cycle set forth

more than 2.0% as measured according to the appropriate test cycle set forth

in the revised NOx Technical Code 2008, or adversely affect engine durability or reliability; and

.2 that the cost of the approved method is not excessive, which is determined by a comparison of the amount of NOx reduced by the approved method to achieve the standard set forth in paragraph 7.4 of this regulation and the cost of purchasing and installing such approved method.

Certification

8 The revised NOx Technical Code 2008 shall be applied in the certification, testing and measurement procedures for the standards set forth in this regulation.

9 The procedures for determining NOx emissions set out in the revised NOx Technical Code 2008 are intended to be representative of the normal operation of the engine. Defeat devices and irrational emission control strategies undermine this intention and shall not be

Code 2008 are intended to be representative of the normal operation of the engine. Defeat devices and irrational emission control strategies undermine this intention and shall not be allowed. This regulation shall not prevent the use of auxiliary control devices that are used to protect the engine and/or its ancillary equipment against operating conditions that could result in damage or failure or that are used to facilitate the starting of the engine.

Regulation 14
Sulphur oxides (SOx) and particulate matter

General requirements

1 The sulphur content of fuel oil used or carried for use on board a ship shall not exceed 0.50% m/m.

2 The worldwide average sulphur content of residual fuel oil supplied for use on board ships shall be monitored taking into account the guidelines developed by the Organization.

Requirements within emission control areas

3 For the purpose of this regulation, an emission control area shall be any sea area, including any port area, designated by the Organization in accordance with the criteria and procedures set forth i n appendix III to this Annex. The emission control areas under this regulation are:

including any port area, designated by the Organization in accordance with the criteria and procedures set forth i n appendix III to this Annex. The emission control areas under this regulation are:

.1 the Baltic Sea area as defined in regulation 1.11.2 of Annex I of the present Convention;

.2 the North Sea area as defined in regulation 1.14.6 of Annex V of the present Convention;

.3 the North American Emission Control Area, which means the area described by the coordinates provided in appendix VII to this Annex; and

.4 the United States Car ibbean Sea Emission Control Area, which means the area described by the coordinates provided in appendix VII to this Annex.

4 While a ship is operating within an emission control area, the sulphur content of fuel oil used on board that ship shall not exceed 0.10% m/m.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 27 -

5 The sulphur content of fuel oil referred to in paragraph 1 and paragraph 4 of this

5 The sulphur content of fuel oil referred to in paragraph 1 and paragraph 4 of this regulation shall be documented by its supplier as required by regulation 18 of this Annex.

6 Those ships using separate fuel oils to comply with paragraph 4 of this regulation and entering or leaving an emission control area set forth in paragraph 3 of this regulation shall carry a written procedure showing how the fuel oil changeover is to be done, allowing sufficient time for the fuel oil service system to be fully flushed of all fuel oils exceeding the applicable sulphur content specified in paragraph 4 of this regulation prior to entry into an emission control area. The volume of low sulphur fuel oils in each tank as well as the date, time and position of the ship when any fuel oil changeover operation is completed prior to the entry into an emission control area or commenced after exit from such an area shall be recorded in such logbook or electronic record book as prescribed by the Administration.

control area or commenced after exit from such an area shall be recorded in such logbook or electronic record book as prescribed by the Administration.

7 During the first 12 months immediately following entry into force of an amendment designating a specific emission control area under paragraph 3 of this regulation, ships operating in that emission control area are exempt from the requirements in paragraphs 4 and 6 of this regulation and from the requirements of paragraph 5 of this regulation insofar as they relate to paragraph 4 of this regulation.

In-use and onboard fuel oil sampling and testing

8 If the competent authority of a Party requires the in -use or onboard sample to be analysed, it shall be done in accordance with the verification procedure set forth in appendix VI to this Annex to determine whether the fuel oil being used or carried for us e on board meets the requirements in paragraph 1 or paragraph 4 of this regulation. The in-use sample shall be drawn taking into account the guidelines developed by the Organization. The onboard sample shall be drawn taking into account the guidelines developed by the Organization.

drawn taking into account the guidelines developed by the Organization. The onboard sample shall be drawn taking into account the guidelines developed by the Organization.

9 The sample shall be sealed by the representative of the competent authority with a unique means of identification installed in the presence of the ship 's representative. The ship shall be given the option of retaining a duplicate sample.

In-use fuel oil sampling point

10 For each ship subject to regulations 5 and 6 of this Annex, sampling point(s) shall be fitted or designated for the purpose of taking representative samples of the fuel oil being used on board the ship taking into account the guidelines developed by the Organization.

11 For a ship constructed before 1 April 2022, the sampling point(s) referred to in paragraph 10 shall be fitted or designated not later than the first renewal survey as identified in regulation 5.1.2 of this Annex on or after 1 April 2023.

12 The requirements of paragraphs 10 and 11 above are not applicable to a fuel oil service system for a low -flashpoint fuel for combustion purposes for propulsion or operation on board the ship.

12 The requirements of paragraphs 10 and 11 above are not applicable to a fuel oil service system for a low -flashpoint fuel for combustion purposes for propulsion or operation on board the ship.

13 The competent authority of a Party shall, as appropriate, utilize the sampling point(s) which is(are) fitted or designated for the purpose of taking representative sample(s) of the fuel oil being used on board in order to verify that the fuel oil complies with this regulation. Taking fuel oil samples by the competent authority of the Party shall be performed as expeditiously as possible without causing the ship to be unduly delayed.

Regulation 15
Volatile organic compounds

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 28 -

1 If the emissions of volatile organic compounds (VOCs) from a tanker are to be regulated in a port or ports or a terminal or terminals under the jurisdiction of a Party, they shall be regulated in accordance with the provisions of this regulation.

regulated in a port or ports or a terminal or terminals under the jurisdiction of a Party, they shall be regulated in accordance with the provisions of this regulation.

2 A Party regulating tank ers for VOC emissions shall submit a notification to the Organization. This notification shall include information on the size of tankers to be controlled, the cargoes requiring vapour emission control systems and the effective date of such control. The notification shall be submitted at least six months before the effective date.

3 A Party that designates ports or terminals at which VOC emissions from tankers are to be regulated shall ensure that vapour emission control systems, approved by that Party taking into account the safety standards for such systems developed by the Organization, are provided in any designated port and terminal and are operated safely and in a manner so as to avoid undue delay to a ship.

4 The Organization shall circulate a list of the ports and terminals designated by Parties to other Parties and Member States of the Organization for their information.

5 A tanker to which paragraph 1 of this regulation applies shall be provided with a

to other Parties and Member States of the Organization for their information.

5 A tanker to which paragraph 1 of this regulation applies shall be provided with a vapour emission collection system approved by the Administration taking into account the safety standards for such systems developed by the Organization, and shall use this system during the loading of relevant cargoes. A port or terminal that has installed vapour emission control systems in accordance with this regulation may accept tankers that are not fitted with vapour collection systems for a period of three years after the effective date identified in paragraph 2 of this regulation.

6 A tanker carrying crude oil shall have on board and implement a VOC management plan approved by the Administration. Such a plan shall be prepared taking into account the guidelines developed by the Organization. The plan shall be specific to each ship and shall at least:

.1 provide written procedures for minimizing VOC emissions during the loading, sea passage and discharge of cargo;

.2 give consideration to the additional VOC generated by crude oil washing;

least:

.1 provide written procedures for minimizing VOC emissions during the loading, sea passage and discharge of cargo;

.2 give consideration to the additional VOC generated by crude oil washing;

.3 identify a person responsible for implementing the plan; and

.4 for ships on international voyages, be written in the working language of the master and officers and, if the working language of the master and officers is not English, French or Spanish, include a translation into one of these languages.

7 This regulation shall also apply to gas carriers only if the types of loading and containment systems allow safe retention of non-methane VOCs on board or their safe return ashore.

Regulation 16
Shipboard incineration

1 Except as provided in paragraph 4 of this regulation, shipboard incineration shall be allowed only in a shipboard incinerator.
2 Shipboard incineration of the following substances shall be prohibited:

.1 residues of cargoes subject to Annex I, II or III or related contaminated packing materials;

.2 polychlorinated biphenyls (PCBs);

.3 garbage, as defined by Annex V, containing more than traces of heavy metals;

.4 refined petroleum products containing halogen compounds;

.5 sewage sludge and sludge oil n either of which is generated on board the ship; and

.6 exhaust gas cleaning system residues.

3 Shipboard incineration of polyvinyl chlorides (PVCs) shall be prohibited, except in shipboard incinerators for which IMO Type Approval Certificates have been issued.

4 Shipboard incineration of sewage sludge and sludge oil generated during normal operation of a ship may also take place in the main or auxiliary power plant or boilers, but in those cases, shall not take place inside ports, harbours or estuaries.

5 Nothing in this regulation either:

.1 affects the incineration at sea prohibitions of the Convention on the

those cases, shall not take place inside ports, harbours or estuaries.

5 Nothing in this regulation either:

.1 affects the incineration at sea prohibitions of the Convention on the Prevention of Marine Pollution by Dumping of Wastes and Other Matter, 1972, as amended, and the 1996 Protocol thereto, or other requirements thereof,
or

.2 precludes the development, installation and operation of alternative design shipboard thermal waste treatment devices that meet or exceed the requirements of this regulation.

6.1 Except as provided in paragraph 6.2 of this regulation, each incinerator on a ship constructed on or after 1 January 2000 or incinerator that is installed on board a ship on or after 1 January 2000 shall meet the requirements contained in appendix IV to this Annex. Each incinerator subject to this paragraph shall be approved by the Administration taking into account the standard specification for shipboard incinerators developed by the Organization;

6.2 The Administration may allow exclusion from the application of paragraph 6.1 of this regulation to any incinerator installed on board a ship before 19 May 2005, provided that the

6.2 The Administration may allow exclusion from the application of paragraph 6.1 of this regulation to any incinerator installed on board a ship before 19 May 2005, provided that the ship is solely engaged in voyages within waters subject to the sovereignty or jurisdiction of the State the flag of which the ship is entitled to fly.

7 Incinerators installed in accordance with the requirements of paragraph 6.1 of this regulation shall be provided with a manufacturer's operating manual, which is to be retained with the unit and which shall specify how to operate the incinerator within the limits described in paragraph 2 of appendix IV of this Annex.

8 Personnel responsible for the operation of an incinerator installed in accordance with the requirements of paragraph 6.1 of this regulation shall be trained to implement the guidance provided in the manufacturer's operating manual as required by paragraph 7 of this regulation.
9 For incinerators installed in accordance with the requirements of paragraph 6.1 of this regulation the combustion chamber gas outlet temperature shall be monitored at all times the

9 For incinerators installed in accordance with the requirements of paragraph 6.1 of this regulation the combustion chamber gas outlet temperature shall be monitored at all times the

unit is in operation. Where that incinerator is of the continuous-feed type, waste shall not be fed into the unit when the combustion chamber gas outlet temperature is below 850°C. Where that incinerator is of the batch-loaded type, the unit shall be designed so that the combustion chamber gas outlet temperature shall reach 600°C within five minutes after start-up and will thereafter stabilize at a temperature not less than 850°C.

Regulation 17
Reception facilities

1 Each Party undertakes to ensure the provision of facilities adequate to meet the:

.1 needs of ships using its repair ports for the reception of ozone-depleting substances and equipment containing such substances when removed from ships;

1 Each Party undertakes to ensure the provision of facilities adequate to meet the:

.1 needs of ships using its repair ports for the reception of ozone-depleting substances and equipment containing such substances when removed from ships;

.2 needs of ships using its ports, terminals or repair ports for the reception of exhaust gas cleaning residues from an exhaust gas cleaning system;

without causing undue delay to ships, and

.3 needs in ship -breaking facilities for the reception of ozone-depleting substances and equipment containing such substances when removed from ships.

2 Small island developing States may satisfy the requirements in paragraph 1 of this regulation through regional arrangements when, because of those States ' unique circumstances, such arrangements are the only practical means to satisfy these requirements. Parties participating in a regional arrangement shall develop a Regional Reception Facilities Plan, taking into account the guidelines developed by the Organization.

The Government of each Party participating in the arrangement shall consult with the Organization for circulation to the Parties of the present Convention:

Plan, taking into account the guidelines developed by the Organization.

The Government of each Party participating in the arrangement shall consult with the Organization for circulation to the Parties of the present Convention:

.1 how the Regional Reception Facilities Plan takes into account the Guidelines;

.2 particulars of the identified Regional Ships Waste Reception Centres; and

.3 particulars of those ports with only limited facilities.

3 If a particular port or terminal of a Party is, taking into account the guidelines to be developed by the Organization, remotely located from, or lacking in, the industrial infrastructure necessary to manage and process those substances referred to in paragraph 1 of this regulation and therefore cannot accept such substances, then the Party shall inform the Organization of any such port or terminal so that this information may be circulated to all Parties and Member States of the Organization for their information and any appropriate action. Each Party that has provided the Organization with such information shall also notify the

and Member States of the Organization for their information and any appropriate action. Each Party that has provided the Organization with such information shall also notify the Organization of its ports and terminals where reception facilities are available to manage and process such substances.

4 Each Party shall notify the Organization for circulation to the Members of the Organization of all cases where the facilities provided under this regulation are unavailable or alleged to be inadequate.

RESOLUTION MEPC.328(76)

AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 31 -

Regulation 18

Fuel oil availability and quality

Fuel oil availability

1 Each Party shall take all reasonable steps to promote the availability of fuel oils that comply with this Annex and inform the Organization of the availability of compliant fuel oils in its ports and terminals.

2.1 If a ship is found by a Party not to be in compliance with the standards for compliant

comply with this Annex and inform the Organization of the availability of compliant fuel oils in its ports and terminals.

2.1 If a ship is found by a Party not to be in compliance with the standards for compliant fuel oils set forth in this Annex, the competent authority of the Party is entitled to require the ship to:

.1 present a record of the actions taken to attempt to achieve compliance; and

.2 provide evidence that it attempted to purchase compliant fuel oil in accordance with its voyage plan and, if it was not made available where planned, that attempts were made to locate alternative sources for such fuel oil and that despite best efforts to obtain compliant fuel oil, no such fuel oil was made available for purchase.

2.2 The ship should not be required to deviate from its intended voyage or to delay unduly the voyage in order to achieve compliance.

2.3 If a ship provides the information set forth in paragraph 2.1 of this regulation, a Party shall take into account all relevant circumstances and the evidence presented to determine the appropriate action to take, including not taking control measures.

shall take into account all relevant circumstances and the evidence presented to determine the appropriate action to take, including not taking control measures.

2.4 A ship shall notify its Administration and the competent authority of the relevant port of destination when it cannot purchase compliant fuel oil.

2.5 A Party shall notify the Organization when a ship has presented evidence of the non-availability of compliant fuel oil.

Fuel oil quality

3 Fuel oil for combustion purposes delivered to and used on board ships to which this Annex applies shall meet the following requirements:

.1 except as provided in paragraph 3.2 of this regulation:

.1.1 the fuel oil shall be blends of hydrocarbons derived from petroleum refining. This shall not preclude the incorporation of small amounts of additives intended to improve some aspects of performance;

.1.2 the fuel oil shall be free from inorganic acid; and

.1.3 the fuel oil shall not include any added substance or chemical waste that:
.1 jeopardizes the safety of ships or adversely affects the performance of the machinery, or

.2 is harmful to personnel, or

.1.3 the fuel oil shall not include any added substance or chemical waste that:
.1 jeopardizes the safety of ships or adversely affects the performance of the machinery, or

.2 is harmful to personnel, or

.3 contributes overall to additional air pollution.

.2 fuel oil for combustion purposes derived by methods other than petroleum refining shall not:

.2.1 exceed the applicable sulphur content set forth in regulation 14 of this Annex;

.2.2 cause an engine to exceed the applicable NOx emission limit set forth in paragraphs 3, 4, 5.1.1 and 7.4 of regulation 13;

.2.3 contain inorganic acid; or

.2.4.1 jeopardize the safety of ships or adversely affect the performance of the machinery, or

.2.4.2 be harmful to personnel, or

.2.4.3 contribute overall to additional air pollution.

4 This regulation does not apply to coal in its solid form or nuclear fuels.

of the machinery, or

.2.4.2 be harmful to personnel, or

.2.4.3 contribute overall to additional air pollution.

4 This regulation does not apply to coal in its solid form or nuclear fuels.
Paragraphs 5, 6, 7.1, 7.2, 8.1, 8.2, 9.2, 9.3, and 9.4 of this regulation do not apply to gas fuels such as liquefied natural gas, compressed natural gas or liquefied petroleum gas. The sulphur content of gas fuels delivered to a ship specifically for combustion purposes on board that ship shall be documented by the supplier.

5 For each ship subject to regulations 5 and 6 of this Annex, details of fuel oil for combustion purposes delivered to and used on board shall be recorded by means of a bunker delivery note that shall contain at least the information specified in appendix V to this Annex.

6 The bunker delivery note shall be kept on board the ship in such a place as to be readily available for inspection at all reasonable times. It shall be retained for a period of three years after the fuel oil has been delivered on board.

7.1 The competent authority of a Party may inspect the bunker delivery notes on board

readily available for inspection at all reasonable times. It shall be retained for a period of three years after the fuel oil has been delivered on board.

7.1 The competent authority of a Party may inspect the bunker delivery notes on board any ship to which this Annex applies while the ship is in its port or offshore terminal, may make a copy of each delivery note, and may require the master or person in charge of the ship to certify that each copy is a true copy of such bunker delivery note. The competent authority may also verify the contents of each note through consultations with the port where the note was issued.

7.2 The inspection of the bunker delivery notes and the taking of certified copies by the competent authority under paragraph 7.1 of this regulation shall be performed as expeditiously as possible without causing the ship to be unduly delayed.

8.1 The bunker delivery note shall be accompanied by a representative sample of the fuel oil delivered taking into account the guidelines developed by the Organization. The sample is to be sealed and signed by the supplier's representative and the master or officer in charge of

oil delivered taking into account the guidelines developed by the Organization. The sample is to be sealed and signed by the supplier's representative and the master or officer in charge of the bunker operation on completion of bunkering operations and retained under the ship's control until the fuel oil is substantially consumed, but in any case for a period of not less than 12 months from the time of delivery.

8.2 If a Party requires the representative sample to be analysed, it shall be done in accordance with the verification procedure set forth in appendix VI to this Annex to determine whether the fuel oil meets the requirements of this Annex.
RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 33 -

9 Parties undertake to ensure that appropriate authorities designated by them:

.1 maintain a register of local suppliers of fuel oil;

.2 require local suppliers to provide the bunker delivery note and sample as required by this regulat ion, certified by the fuel oil supplier that the fuel oil

.1 maintain a register of local suppliers of fuel oil;

.2 require local suppliers to provide the bunker delivery note and sample as required by this regulat ion, certified by the fuel oil supplier that the fuel oil meets the requirements of regulations 14 and 18 of this Annex;

.3 require local suppliers to retain a copy of the bunker delivery note for at least three years for inspection and verification by the port State as necessary;

.4 take action as appropriate against fuel oil suppliers that have been found to deliver fuel oil that does not comply with that stated on the bunker delivery note;

.5 inform the Administration of any ship receiving fuel oil found to be non-compliant with the requirements of regulation 14 or 18 of this Annex; and

.6 inform the Organization for circulation to Parties and Member States of the Organization of all cases where fuel oil suppliers have fa iled to meet the requirements specified in regulations 14 or 18 of this Annex.

10 In connection with port State inspections carried out by Parties, the Parties further undertake to:

.1 inform the Party or non-Party under whose jurisdiction a bunker delivery note

10 In connection with port State inspections carried out by Parties, the Parties further undertake to:

.1 inform the Party or non-Party under whose jurisdiction a bunker delivery note was issued of cases of delivery of non-compliant fuel oil, giving all relevant information; and

.2 ensure that remedial action as appropriate is taken to bring non-compliant fuel oil discovered into compliance.

11 For every ship of 400 gross tonnage and above on scheduled services with frequent and regular port calls, an Administration may decide after application and consultation with affected States that compliance with paragraph 6 of this regulation may be documented in an alternative manner that gives similar certainty of compliance with regulations 14 and 18 of this Annex.

Chapter 4 Regulations on the carbon intensity of international shipping

Regulation 19
Application

1 This chapter shall apply to all ships of 400 gross tonnage and above.

2 The provisions of this chapter shall not apply to:

.1 ships solely engaged in voyages within waters subject to the sovereignty or jurisdiction of the State the flag of which the ship is entitled to fly. However,

2 The provisions of this chapter shall not apply to:

.1 ships solely engaged in voyages within waters subject to the sovereignty or jurisdiction of the State the flag of which the ship is entitled to fly. However, each Party should ensure, by the adoption of appropriate measures, that such ships are constructed and act in a manner consistent with the requirements of chapter 4 of this Annex, so far as is reasonable and practicable.

.2 ships not propelled by mechanical means, and platforms including FPSOs and FSUs and drilling rigs, regardless of their propulsion.

3 Regulations 22, 23, 24 and 25 of this Annex shall not apply to ships which have non-conventional propulsion, except that regulations 22 and 24 shall apply to cruise passenger ships having non -conventional propulsion and LNG carriers having conventional or non-conventional propulsion, delivered on or after 1 September 2019, as defined in

ships having non -conventional propulsion and LNG carriers having conventional or non-conventional propulsion, delivered on or after 1 September 2019, as defined in regulation 2.2.1, and regulations 23 and 25 shall apply to cruise passenger ships having non-conventional propulsion and LNG carriers having conventional or non -conventional propulsion. Regulations 22, 23, 24, 25 and 28 shall not apply to category A ships as defined in the Polar Code.

4 Notwithstanding the provisions of paragraph 1 of this regulation, the Administration may waive the requirement for a ship of 400 gross tonnage and above to comply with regulations 22 and 24 of this Annex.

5 The provision of paragraph 4 of this regulation shall not apply to ships of 400 gross tonnage and above:

.1 for which the building contract is placed on or after 1 January 2017; or

.2 in the absence of a building contract, the keel of which is laid or which is at a similar stage of construction on or after 1 July 2017; or

.3 the delivery of which is on or after 1 July 2019; or

.4 in cases of a major conversion of a new or existing ship, as defined in

a similar stage of construction on or after 1 July 2017; or

.3 the delivery of which is on or after 1 July 2019; or

.4 in cases of a major conversion of a new or existing ship, as defined in regulation 2.2.17 of this Annex, on or after 1 Janu ary 2017, and in which regulations 5.4.2 and 5.4.3 of this Annex apply.

6 The Administration of a Party to the present Convention which allows the application of paragraph 4, or suspends, withdraws or declines the application of that paragraph, to a ship entitled to fly its flag shall forthwith communicate to the Organization for circulation to the Parties to the present Protocol particulars thereof, for their information.

Regulation 20
Goal

The goal of this chapter is to reduce the carbon intensity of international shipping, working towards the levels of ambition set out in the Initial IMO Strategy on reduction of GHG emissions from ships.

Regulation 21
Functional requirements

In order to achieve the goal set out in regulation 20 of this Annex, a ship to which this chapter applies shall comply, as applicable, with the following functional requirements to reduce its carbon intensity:

Functional requirements

In order to achieve the goal set out in regulation 20 of this Annex, a ship to which this chapter applies shall comply, as applicable, with the following functional requirements to reduce its carbon intensity:
.1 the technical carbon intensity requirements in accorda nce with regulations 22, 23, 24 and 25 of this Annex; and

.2 the operational carbon intensity requirements in accordance with regulations 26, 27 and 28 of this Annex.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 35 -

Regulation 22
Attained Energy Efficiency Design Index (attained EEDI)

1 The attained EEDI shall be calculated for:

.1 each new ship;

.2 each new ship which has undergone a major conversion; and

.3 each new or existing ship which has undergone a major conversion that is so extensive that the ship is regarded by the Administration as a newly constructed ship,

which falls into one or more of the categories in regulations 2.2.5, 2.2.7, 2.2.9, 2.2.11, 2.2.14

so extensive that the ship is regarded by the Administration as a newly constructed ship,

which falls into one or more of the categories in regulations 2.2.5, 2.2.7, 2.2.9, 2.2.11, 2.2.14 to 2.2.16, 2.2.20, 2.2.22, and 2.2.26 to 2.2.29 of this Annex. The attained EEDI shall be specific to each ship and shall indicate the estimated performance of the ship in terms of energy efficiency, and be accompanied by the EEDI technical file that contains the information necessary for the calculation of the attained EEDI and that shows the process of calculation. The attained EEDI shall be verified, based on the EEDI technical file, either by the Administration or by any organization duly authorized by it.

2 The attained EEDI shall be calculated taking into account the guidelines develo ped by the Organization.

3 For each ship subject to regulation 24 of this Annex, the Administration or any organization duly authorized by it shall report to the Organization the required and attained EEDI values and relevant information, taking into acco unt the guidelines developed by the Organization, via electronic communication:

organization duly authorized by it shall report to the Organization the required and attained EEDI values and relevant information, taking into acco unt the guidelines developed by the Organization, via electronic communication:

.1 within seven months of completing the survey required under regulation 5.4 of this Annex; or

.2 within seven months following 1 April 2022 for a ship delivered prior to 1 April 2022.

Regulation 23
Attained Energy Efficiency Existing Ship Index (attained EEXI)

1 The attained EEXI shall be calculated for:

.1 each ship; and

.2 each ship which has undergone a major conversion,

which falls into one or more of the categories in regulations 2.2.5, 2.2.7, 2.2.9, 2.2.11, 2.2.14 to 2.2.16, 2.2.22, and 2.2.26 to 2.2.29 of this Annex. The attained EEXI shall be specific to each ship and shall indicate the estimated performance of the sh ip in terms of energy efficiency, and be accompanied by the EEXI technical file which contains the information necessary for the calculation of the attained EEXI and which shows the process of the calculation. The attained EEXI shall be verified, based on the EEXI technical file, either by the

necessary for the calculation of the attained EEXI and which shows the process of the calculation. The attained EEXI shall be verified, based on the EEXI technical file, either by the Administration or by any organization duly authorized by it.

2 The attained EEXI shall be calculated taking into account the guidelines developed by the Organization.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 36 -

3 Notwithstanding paragraph 1 of this regulation , for each ship to which regulation 22 of this Annex applies, the attained EEDI verified by the Administration or by any organization duly authorized by it in accordance with regulation 22.1 of this Annex may be taken as the attained EEXI if the value of the attained EEDI is equal to or less than that of the required EEXI as required by regulation 25 of this Annex. In this case, the attained EEXI shall be verified based on the EEDI technical file.

Regulation 24
Required EEDI

1 For each:

.1 new ship,

as required by regulation 25 of this Annex. In this case, the attained EEXI shall be verified based on the EEDI technical file.

Regulation 24
Required EEDI

1 For each:

.1 new ship,

.2 new ship which has undergone a major conversion, and

.3 new or existing ship which has undergone a major conversion that is so extensive that the ship is regarded by the Administration as a newly constructed ship

which falls into one of the categories in regulations 2.2.5, 2.2.7, 2.2.9, 2.2.11, 2.2.14 to 2.2.16, 2.2.22, and 2.2.26 to 2.2.29 and to which this chapter is applicable, the attained EEDI shall be as follows:

Attained EEDI Required EEDI = (1
X
100) Reference line value

where X is the reduction factor specified in table 1 for the required EEDI compared to the EEDI reference line.

2 For each new and existing ship that has undergone a major conversion which is so extensive that the ship is regarded by the Administration as a newly constructed ship, the attained EEDI shall be calculated and meet the requirement of paragraph 1 of this regulation with the reduction factor applicable corresponding to the ship type and size of the converted

attained EEDI shall be calculated and meet the requirement of paragraph 1 of this regulation with the reduction factor applicable corresponding to the ship type and size of the converted ship at the date of the contract of the conversion, or in the absence of a contract, the commencement date of the conversion.

Table 1 - Reduction factors (in percentage) for the EEDI relative to the EEDI reference line

Ship Type Size
Phase 0
1 Jan
2013
31 Dec
2014
Phase 1
1 Jan
2015
31 Dec
2019
Phase 2
1 Jan
2020
31 Mar
2022
Phase 2
1 Jan
2020
31 Dec
2024
Phase 3
1 Apr
2022
and
onwards
Phase 3
1 Jan
2025
and
onwards
Bulk carrier
20,000 DWT
and above 0 10 20 30
10,000 and
above but less
than 20,000
DWT
n/a 0-10*

0-20* 0-30*
Gas carrier
15,000 DWT
and above 0 10 20 30
10,000 and
above but less

than 15,000
DWT
0 10

20 30
2,000 and
above but less
than 10,000
DWT
n/a 0-10*

0-20* 0-30*
Tanker

than 20,000
DWT
n/a 0-10*

0-20* 0-30*
Gas carrier
15,000 DWT
and above 0 10 20 30
10,000 and
above but less
than 15,000
DWT
0 10

20 30
2,000 and
above but less
than 10,000
DWT
n/a 0-10*

0-20* 0-30*
Tanker
20,000 DWT
and above 0 10 20 30
4,000 and
above but less
than 20,000
DWT
n/a 0-10*

0-20* 0-30*
Containership
200,000 DWT
and above 0 10 20 50
120,000 and
above but less
than 200,000
DWT
0 10
20
45
80,000 and
above but less
than 120,000
DWT
0 10 20 40
40,000 and
above but less
than 80,000
DWT
0 10 20 35
15,000 and
above but less
than 40,000
DWT
0 10 20 30
10,000 and

above but less
than 15,000
DWT
n/a 0-10* 0-20* 15-30*
General
Cargo ships
15,000 DWT
and above 0 10 15 30
RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 38 -

Ship Type Size
Phase 0
1 Jan
2013
31 Dec
2014
Phase 1
1 Jan
2015
31 Dec
2019
Phase 2
1 Jan
2020
31 Mar
2022
Phase 2
1 Jan
2020
31 Dec
2024
Phase 3
1 Apr
2022
and
onwards

Ship Type Size
Phase 0
1 Jan
2013
31 Dec
2014
Phase 1
1 Jan
2015
31 Dec
2019
Phase 2
1 Jan
2020
31 Mar
2022
Phase 2
1 Jan
2020
31 Dec
2024
Phase 3
1 Apr
2022
and
onwards
Phase 3
1 Jan
2025
and
onwards
3,000 and
above but less
than 15,000
DWT
n/a 0-10* 0-15* 0-30*
Refrigerated
cargo carrier
5,000 DWT
and above 0 10 15 30
3,000 and
above but less
than 5,000
DWT
n/a 0-10*

0-15* 0-30*
Combination
carrier
20,000 DWT
and above 0 10 20 30
4,000 and
above but less
than 20,000
DWT
n/a 0-10*

0-20* 0-30*

LNG carrier*** 10,000 DWT and above n/a 10** 20 30
Ro-ro cargo ship (vehicle carrier)*** 10,000 DWT and above n/a 5** 15 30
Ro-ro cargo ship*** 2,000 DWT and above n/a 5** 20 30
1,000 and above but less than 2,000 DWT n/a 0-5*,** 0-20* 0-30*
Ro-ro passenger ship*** 1,000 DWT and above n/a 5** 20 30
250 and above but less than 1,000 DWT n/a 0-5*,** 0-20* 0-30*
Cruise passenger ship*** having non-conventional propulsion 85,000 GT and above n/a 5** 20 30
25,000 and above but less than 85,000 GT n/a 0-5*,** 0-20* 0-30*

250 and above
but less than
1,000 DWT
n/a 0-5*,** 0-20* 0-30*
Cruise
passenger
ship***
having
non-
conventional
propulsion
85,000 GT
and above n/a 5** 20 30
25,000 and
above but less
than 85,000
GT
n/a 0-5*,** 0-20* 0-30*
* Reduction factor to be linearly interpolated between the two values dependent upon ship size. The lower value of the reduction factor is to be applied to the smaller ship size.
** Phase 1 commences for those ships on 1 September 2015.
*** Reduction factor applies to those ships delivered on or after 1 September 2019, as defined in paragraph 2.1 of regulation 2.
Note: n/a means that no required EEDI applies.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 39 -

3 The reference line values shall be calculated as follows:

Reference line value = $a \cdot bc$

where a, b and c are the parameters given in table 2.

Table 2 - Parameters for the determination of reference values for the different ship types

Ship type defined in regulation 2 a b c

Reference line value = a · bc

where a, b and c are the parameters given in table 2.

Table 2 - Parameters for the determination of reference values for the different ship types

Ship type defined in regulation 2 a b c
2.2.5 Bulk carrier 961.79 DWT of the ship
where
DWT279,000

279,000 where
DWT > 279,000
0.477
2.2.7 Combination carrier 1,219.00 DWT of the ship 0.488
2.2.9 Containership 174.22 DWT of the ship 0.201
2.2.11 Cruise passenger ship
having non-conventional
propulsion
170.84 GT of the ship 0.214
2.2.14 Gas carrier 1,120.00 DWT of the ship 0.456
2.2.15 General cargo ship 107.48 DWT of the ship 0.216
2.2.16 LNG carrier 2,253.7 DWT of the ship 0.474
2.2.22 Refrigerated cargo carrier 227.01 DWT of the ship 0.244
2.2.26 Ro-ro cargo ship 1405.15 DWT of the ship

0.498
1686.17* DWT of the ship
where
DWT17,000*

17,000 where
DWT > 17,000*
2.2.27 Ro -ro cargo ship (vehicle
carrier)
(DWT/GT)-
0.7 · 780.36
where DWT/GT < 0.3
1,812.63
where DWT/GT 0.3
DWT of the ship

0.471
2.2.28 Ro-ro passenger ship 752.16 DWT of the ship

0.381
902.59* DWT of the ship
where
DWT10,000*

10,000 where
DWT > 10,000*

(DWT/GT)-
0.7 · 780.36
where DWT/GT < 0.3
1,812.63
where DWT/GT 0.3
DWT of the ship

0.471
2.2.28 Ro-ro passenger ship 752.16 DWT of the ship

0.381
902.59* DWT of the ship
where
DWT10,000*

10,000 where
DWT > 10,000*
2.2.29 Tanker 1,218.80 DWT of the ship 0.488
* to be used from phase 2 and thereafter.

4 If the design of a ship allows it to fall into more than one of the ship type definitions specified in table 2, the required EEDI for the ship shall be the most stringent (the lowest) required EEDI.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 40 -

5 For each ship to which this regulation applies, the installed propulsion power shall not be less than the propulsion power needed to maintain the manoeuvrability of the ship under adverse conditions as defined in the guidelines to be developed by the Organization.

6 At the beginning of phase 1 and at the midpoint of phase 2, the Organization shall

adverse conditions as defined in the guidelines to be developed by the Organization.

6 At the beginning of phase 1 and at the midpoint of phase 2, the Organization shall review the status of technological developments and, if proven necessary, amend the time periods, the EEDI reference line parameters for relevant ship types and reduction rates set out in this regulation.

Regulation 25
Required EEXI

1 For:

.1 each ship; and

.2 each ship which has undergone a major conversion

which falls into one of the categories in regulations 2.2.5, 2.2.7, 2.2.9, 2.2.11, 2.2.14 to 2.2.16, 2.2.22, and 2.2.26 to 2.2.29 and to which this chapter is applicable, the attained EEXI shall be as follows:

Attained EEXI Required EEXI = (1
Y
100) EEDI reference line value

where Y is the reduction factor specified in Table 3 for the required EEXI compared to the EEDI reference line.

Table 3 - Reduction factors (in percentage) for the EEXI relative to the EEDI reference line

Ship type	Size	Reduction factor
Bulk carrier	200,000 DWT and above	15
	20,000 and above but less than 200,000 DWT	20
	10,000 and above but less than 20,000 DWT	0-20*
Gas carrier		

line

Ship type	Size	Reduction factor
Bulk carrier		
	200,000 DWT and above	15
	20,000 and above but less than 200,000 DWT	20
	10,000 and above but less than 20,000 DWT	0-20*
Gas carrier		
	15,000 DWT and above	30
	10,000 and above but less than 15,000 DWT	20
	2,000 and above but less than 10,000 DWT	0-20*
Tanker		
	200,000 DWT and above	15
	20,000 and above but less than 200,000 DWT	20
	4,000 and above but less than 20,000 DWT	0-20*
Containership	200,000 DWT and above	50

Ship type	Size	Reduction factor
	120,000 and above but less than 200,000 DWT	45
	80,000 and above but less than 120,000 DWT	35
	40,000 and above but less than 80,000 DWT	30
	15,000 and above but less than 40,000 DWT	20
	10,000 and above but less than 15,000 DWT	0-20*
General cargo ship		
	15,000 DWT and above	30
	3,000 and above but less than 15,000 DWT	0-30*
Refrigerated cargo carrier		
	5,000 DWT and above	15
	3,000 and above but less than 5,000 DWT	0-15*

10,000 and above but less
than 15,000 DWT 0-20*
General cargo ship
15,000 DWT and
above 30
3,000 and above but less
than 15,000 DWT 0-30*
Refrigerated cargo carrier
5,000 DWT and
above 15
3,000 and above but less
than 5,000 DWT 0-15*
Combination carrier
20,000 DWT and
above 20
4,000 and above but less
than 20,000 DWT 0-20*
LNG carrier 10,000 DWT and above 30
Ro-ro cargo ship (vehicle
carrier)
10,000 DWT and
above 15
Ro-ro cargo ship
2,000 DWT and
above 5
1,000 and above but less
than 2,000 DWT 0-5*
Ro-ro passenger ship
1,000 DWT and
above 5
250 and above but less
than 1,000 DWT 0-5*
Cruise passenger ship
having non-conventional
propulsion
85,000 GT
and above 30
25,000 and above but less
than 85,000 GT 0-30*

* Reduction factor to be linearly interpolated between the two values dependent upon ship size.

The lower value of the reduction factor is to be applied to the smaller ship size.

2 The EEDI reference line values shall be calculated in accordance with regulations 24.3 and 24.4 of this Annex. For ro-ro cargo ships and ro-ro passenger ships, the

2 The EEDI reference line values shall be calculated in accordance with regulations 24.3 and 24.4 of this Annex. For ro-ro cargo ships and ro-ro passenger ships, the reference line value to be used from phase 2 and thereafter under regulation 24.3 of this Annex shall be referred to.

3 A review shall be completed by 1 January 2026 by the Organization to assess the effectiveness of this regulation taking into account any guidelines developed by the Organization. If, based on the review, the Parties decide to adopt amendments to this

regulation, such amendments shall be adopted and brought into force in accordance with the provisions of article 16 of the present Convention.

Regulation 26
Ship Energy Efficiency Management Plan (SEEMP)

1 Each ship shall keep on board a ship specific Ship Energy Efficiency Management Plan (SEEMP). This may form part of the ship 's Safety Management System (SMS).

Regulation 26
Ship Energy Efficiency Management Plan (SEEMP)

1 Each ship shall keep on board a ship specific Ship Energy Efficiency Management Plan (SEEMP). This may form part of the ship 's Safety Management System (SMS). The SEEMP shall be developed and reviewed, taking into account the guidelines adopted by the Organization.

2 In the case of a ship of 5,000 gross tonnage and above, the SEEMP shall include a description of the methodology that will be used to collect the data required by regulation 27.1 of this Annex and the processes that will be used to report the data to the ship's Administration.

3 In the case of a ship of 5,000 gross tonnage and above, which falls into one or more of the categories in regulations 2.2.5, 2.2.7, 2.2.9, 2.2.11, 2.2.14 to 2.2.16, 2.2.22, and 2.2.26 to 2.2.29 of this Annex:

.1 On or before 1 January 2023 the SEEMP shall include:

.1 a description of the methodology that will be used to calculate the ship's attained annual o perational CII required by regulation 28 of this Annex and the processes that will be used to report this value to the ship's Administration;

.1 a description of the methodology that will be used to calculate the ship's attained annual o perational CII required by regulation 28 of this Annex and the processes that will be used to report this value to the ship's Administration;

.2 the required annual operational CII, as specified in regulation 28 of this Annex, for the next three years;

.3 an implementation plan documenting how the required annual operational CII will be achieved during the next three years; and

.4 a procedure for self-evaluation and improvement.

.2 For a ship rated as D for three consecutive years or rated as E in accordance with regulation 28 of this Annex, the SEEMP shall be reviewed in accordance with regulation 28.8 of this Annex to include a plan of corrective actions to achieve the required annual operational CII.

.3 The SEEMP shall be subject to verification and company audits taking into account the guidelines to be developed by the Organization.

Regulation 27
Collection and reporting of ship fuel oil consumption data

1 From calendar year 2019, each ship of 5,000 gross tonnage and above shall collect

account the guidelines to be developed by the Organization.

Regulation 27
Collection and reporting of ship fuel oil consumption data

1 From calendar year 2019, each ship of 5,000 gross tonnage and above shall collect the data specified in appendix IX to this Annex, for that and each subsequent calendar year or portion thereof, as appropriate according to the methodology included in the SEEMP.

2 Except as provided for in paragraphs 4, 5 and 6 of this regulation, at the end of each calendar year, the ship shall aggregate the data collected in that calendar year or portion thereof, as appropriate.

3 Except as provided for in paragraphs 4, 5 and 6 of this regulation, within three months after the end of each calendar year, the ship shall report to its Administration or any

organization duly authorized by it, the aggregated value for each datum specified in appendix

organization duly authorized by it, the aggregated value for each datum specified in appendix IX to this Annex, via electronic communication and using a standardized format to be developed by the Organization.

4 In the event of the transfer of a ship from one Administration to another, the ship shall on the day of completion of the transfer or as close as practical thereto report to the losing Administration or any organization duly authorized by it 41, the aggregated data for the period of the calendar year corresponding to that Administration, as specified in appendix IX to this Annex and, upon prior request of that Administration, the disaggregated data.

5 In the event of a change from one company to another, the ship shall on the day of completion of the change or as close as practical thereto report to its Administration or any organization duly authorized by it, the aggregated data for the portion of the calendar year corresponding to the company, as specified in appendix IX to this Annex and, upon request of

organization duly authorized by it, the aggregated data for the portion of the calendar year corresponding to the company, as specified in appendix IX to this Annex and, upon request of its Administration, the disaggregated data.

6 In the event of change from one Administration to another and from one company to another concurrently, paragraph 4 of this regulation shall apply.

7 The data shall be verified according to procedures established by the Administration, taking into account the guidelines developed by the Organization.

8 Except as provided for in paragraphs 4, 5 and 6 of this regulation, the disaggregated data that underlies the reported data noted in appendix IX to this Annex for the previous calendar year shall be readily accessible for a period of not less than 12 months from the end of that calendar year and be made available to the Administration upon request.

9 The Administration shall ensure that the reported data noted in appendix IX to this Annex by its registered ships of 5,000 gross tonnage and above are transferred to the IMO Ship Fuel Oil Consumption Database via electronic communication and using a standardized

Annex by its registered ships of 5,000 gross tonnage and above are transferred to the IMO Ship Fuel Oil Consumption Database via electronic communication and using a standardized format to be developed by the Organization not later than one month after issuing the Statements of Compliance of these ships.

10 On the basis of the reported data submitted to the IMO Ship Fuel Oil Consumption Database, the Secretary -General of the Organization shall produce an annual report to the Marine Environment Protection Committee summarizing the data collected, the status of missing data, and such other relevant information as may be requested by the Committee.

11 The Secretary-General of the Organization shall grant the Administration of a ship to which regulation 28 of this Annex applies access to all the reported data for all the pre ceding calendar year in the IMO Ship Fuel Oil Consumption Database for that ship.

12 The Secretary-General of the Organization shall maintain an anonymized database such that identification of a specific ship will not be possible. Parties shall have acces s to the anonymized data strictly for their analysis and consideration.

such that identification of a specific ship will not be possible. Parties shall have acces s to the anonymized data strictly for their analysis and consideration.

13 The IMO Ship Fuel Oil Consumption Database shall be undertaken and managed by the Secretary -General of the Organization, pursuant to guidelines to be developed by the Organization.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 44 -

Regulation 28
Operational carbon intensity

Attained annual operational carbon intensity indicator (attained annual operational CII)

1 After the end of calendar year 2023 and after the end of each following calendar year, each ship of 5,000 gross tonnage and above which falls into one or more of the categories in regulations 2.2.5, 2.2.7, 2.2.9, 2.2.11, 2.2.14 to 2.2.16, 2.2.22, and 2.2.26 to 2.2.29 of this Annex shall calculate the attained annual operational CII over a 12 -month period from 1 January to 31 December for the preceding calendar year, using the data c ollected in

Annex shall calculate the attained annual operational CII over a 12 -month period from 1 January to 31 December for the preceding calendar year, using the data c ollected in accordance with regulation 27 of this Annex, taking into account the guidelines to be developed by the Organization.

2 Within three months after the end of each calendar year, the ship shall report to its Administration, or any organization du ly authorized by it, the attained annual operational CII via electronic communication and using a standardized format to be developed by the Organization.

3 Notwithstanding 1 and 2 of this regulation, in the event of any transfer of a ship addressed in regulations 27.4, 27.5 or 27.6 completed after 1 January 2023, a ship shall, after the end of the calendar year in which the transfer takes place, calculate and report the attained annual operational CII for the full 12 -month period from 1 Jan uary to 31 December in the calendar year during which the transfer took place, in accordance with regulations 28.1 and 28.2, for verification in accordance with regulation 6.6 of this Annex, taking into account

calendar year during which the transfer took place, in accordance with regulations 28.1 and 28.2, for verification in accordance with regulation 6.6 of this Annex, taking into account guidelines to be developed by the Organization. Nothing in this regulation relieves any ship of its reporting obligations under regulation 27 or this regulation of this Annex.

Required annual operational carbon intensity indicator (required annual operational CII)

4 For each ship of 5,000 gross tonnage and above which falls into one or more of the categories in regulations 2.2.5, 2.2.7, 2.2.9, 2.2.11, 2.2.14 to 2.2.16, 2.2.22, and 2.2.26 to 2.2.29 of this Annex, the required annual operational CII shall be determined as follows:

Required annual operational CII = (1 Z100) CIIR
where,

Z is the annual reduction factor to ensure continuous improvement of the ship 's operational carbon intensity within a specific rating level; and

CIIR is the reference value.

5 The annual reduction factor Z and the reference value CIIR shall be the values defined taking into account the guidelines to be developed by the Organization.
Operational carbon intensity rating

CIIR is the reference value.

5 The annual reduction factor Z and the reference value CIIR shall be the values defined taking into account the guidelines to be developed by the Organization.
Operational carbon intensity rating

6 The attained annual operational CII shall be documented and verified against the required annual operational CII to determine operational carbon intensity rating A, B, C, D or E, indicating a major superior, minor superior, moderate, minor inferior, or inferior performance level, either by the Administration or by any organization duly authorized by it, taking into account the guidelines developed by the Organization. The middle point of rating level C shall be the value equivalent to the required annual operational CII set ou t in paragraph 4 of this regulation.

Corrective actions and incentives
RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 45 -

7 A ship rated as D for three consecutive years or rated as E shall develop a plan of

7 A ship rated as D for three consecutive years or rated as E shall develop a plan of corrective actions to achieve the required annual operational CII.

8 The SEEMP shall be reviewed to include the plan of corrective actions accordingly , taking into account the guidelines to be developed by the Organization . The revised SEEMP shall be submitted to the Administration or any organization duly authorized by it for verification, preferably together with, but in no case later than 1 month after reporting the attained annual operational CII in accordance with paragraph 2 of this regulation.

9 A ship rated as D for three consecutive years or rated as E shall duly undertake the planned corrective actions in accordance with the revised SEEMP.

10 Administrations, port authorities and other stakeholders as appropriate, are encouraged to provide incentives to ships rated as A or B.

Review

11 A review shall be completed by 1 January 2026 by the Organization to assess:

10 Administrations, port authorities and other stakeholders as appropriate, are encouraged to provide incentives to ships rated as A or B.

Review

11 A review shall be completed by 1 January 2026 by the Organization to assess:

.1 the effectiveness of this regulation in reducing the carbon intensity of international shipping;

.2 the need for reinforced corrective actions or other means of remedy, including possible additional EEXI requirements;

.3 the need for enhancement of the enforcement mechanism;

.4 the need for enhancement of the data collection system; and

.5 the revision of the Z factor and CIIR values.

If based on the review the Parties decide to adopt amendments to this regulation, such amendments shall be adopted and brought into force in accordance with the provisions of article 16 of the present Convention.

Regulation 29
Promotion of technical cooperation and transfer of technology relating to the improvement of energy efficiency of ships

1 Administrations shall, in cooperation with the Organization and other international bodies, promote and provide support, as appropriate, directly or through the Organization to

energy efficiency of ships

1 Administrations shall, in cooperation with the Organization and other international bodies, promote and provide support, as appropriate, directly or through the Organization to States that request technical assistance, especially developing States.

2 The Administration of a Party shall cooperate actively with other Parties, subject to its national laws, regulations and policies, to promote the development and transfer of technology and exchange of information to States which request technical assistance, particularly developing States, in respect of the implementation of measures to fulfil the requirements of chapter 4 of this Annex, in particular regulations 19.4 to 19.6.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 46 -

Chapter 5 Verification of compliance with the provisions of this Annex

Regulation 30
Application

Parties shall use the provisions of the Code for Implementation in the execution of their

Chapter 5 Verification of compliance with the provisions of this Annex

Regulation 30
Application

Parties shall use the provisions of the Code for Implementation in the execution of their obligations and responsibilities contained in this Annex.

Regulation 31
Verification of compliance

1 Every Party shall be subject to periodic audits by the Organization in accordance with the audit standard to verify compliance with and implementation of this Annex.

2 The Secretary-General of the Organization shall have responsibility for administering the Audit Scheme, based on the guidelines developed by the Organization.

3 Every Party shall have responsibility for facilitating the conduct of the audit and implementation of a programme of actions to address the findings, based on the guidelines developed by the Organization.

4 The audits of all Parties shall be:

.1 based on an overall schedule developed by the Secretary-General of the Organization, taking into account the guidelines developed by the Organization; and

.2 conducted at periodic intervals, taking into account the guidelines developed by the Organization.

Organization, taking into account the guidelines developed by the Organization; and

.2 conducted at periodic intervals, taking into account the guidelines developed by the Organization.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 47 -

Appendix I

Form of International Air Pollution Prevention (IAPP) Certificate (regulation 8)

INTERNATIONAL AIR POLLUTION PREVENTION CERTIFICATE

Issued under the provisions of the Protocol of 1997, as amended, to amend the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto (hereinafter referred to as "the Convention") under the authority of the Government of:
..
(full designation of the country)
by..
(full designation of the competent person or organization
authorized under the provisions of the Convention)

Particulars of ship

Name of ship ...

by..
(full designation of the competent person or organization
authorized under the provisions of the Convention)

Particulars of ship

Name of ship ...

Distinctive number or letters ....

IMO number ..

Port of registry ....

Gross tonnage ....

THIS IS TO CERTIFY:

1 That the ship has been surveyed in accordance with regulation 5 of Annex VI of the Convention; and

2 That the survey shows that the equipment, systems, fittings, arrangements and materials fully comply with the applicable requirements of Annex VI of the Convention.

This Certificate is valid until (dd/mm/yyyy) .
subject to surveys in accordance with regulation 5 of Annex VI of the Convention.

Completion date of the survey on which this Certificate is based (dd/mm/yyyy)

Issued at ..
(place of issue of Certificate)

Date (dd/mm/yyyy) ..
(date of issue) (signature of duly authorized

Issued at ..
(place of issue of Certificate)

Date (dd/mm/yyyy) ..
(date of issue) (signature of duly authorized
official issuing the Certificate)

(seal or stamp of the authority, as appropriate)

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 48 -

ENDORSEMENT FOR ANNUAL AND INTERMEDIATE SURVEYS

THIS IS TO CERTIFY that, at a survey required by regulation 5 of Annex VI of the Convention, the ship was found to comply with the relevant provisions of that Annex:

Annual survey Signed. .
(signature of duly authorized official)

Place . .

Date (dd/mm/yyyy)
(seal or stamp of the authority, as appropriate)

Annual/Intermediate survey Signed
(signature of duly authorized official)

Place ..

Date (dd/mm/yyyy) ..
(seal or stamp of the authority, as appropriate)

Annual/Intermediate survey Signed

(signature of duly authorized official)

Place ..

Date (dd/mm/yyyy) ..
(seal or stamp of the authority, as appropriate)

Annual/Intermediate survey Signed
(signature of duly authorized official)

Place ..

Date (dd/mm/yyyy)
(seal or stamp of the authority, as appropriate)

Annual survey Signed. .
(signature of duly authorized official)

Place..

Date (dd/mm/yyyy)
(seal or stamp of the authority, as appropriate)

ANNUAL/INTERMEDIATE SURVEY IN ACCORDANCE
WITH REGULATION 9.8.3

THIS IS TO CERTIFY that, at an annual/intermediate survey in a ccordance with regulation 9.8.3 of Annex VI of the Convention, the ship was found to comply with the relevant provisions of that Annex:

Signed
(signature of duly authorized official)

Place ..

Date (dd/mm/yyyy) ..

(seal or stamp of the authority, as appropriate)

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS,

(seal or stamp of the authority, as appropriate)

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 49 -

ENDORSEMENT TO EXTEND THE CERTIFICATE IF VALID FOR LESS THAN FIVE YEARS WHERE REGULATION 9.3 APPLIES

The ship complies with the relevant provisions of the Annex, and this Cert ificate shall, in accordance with regulation 9.3 of Annex VI of the Convention, be accepted as valid until (dd/mm/yyyy) ...

Signed.
(signature of duly authorized official)

Place .

Date (dd/mm/yyyy)

(seal or stamp of the authority, as appropriate)

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 50 -

ENDORSEMENT WHERE THE RENEWAL SURVEY HAS BEEN COMPLETED AND REGULATION 9.4 APPLIES

The ship complies with the relevant provisions of the Annex, and this Certificate shall, in

1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 50 -

ENDORSEMENT WHERE THE RENEWAL SURVEY HAS BEEN
COMPLETED AND REGULATION 9.4 APPLIES

The ship complies with the relevant provisions of the Annex, and this Certificate shall, in accordance with regulation 9.4 of Annex VI of the Convention, be accepted as valid until (dd/mm/yyyy)

Signed. .
(signature of duly authorized official)

Place .

Date (dd/mm/yyyy)

(seal or stamp of the authority, as appropriate)

ENDORSEMENT TO EXTEND THE VALIDITY OF THE CERTIFICATE
UNTIL REACHING THE PORT OF SURVEY OR FOR A PERIOD OF GRACE
WHERE REGULATION 9.5 OR 9.6 APPLIES

This Certificate shall, in accordance with regulation 9.5 or 9.6 of Annex VI of the Convention, be accepted as valid until (dd/mm/yyyy)..

Signed. ...
(signature of duly authorized official)

Place

Date (dd/mm/yyyy) .

(seal or stamp of the authority, as appropriate)

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE

(signature of duly authorized official)

Place

Date (dd/mm/yyyy) .

(seal or stamp of the authority, as appropriate)

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 51 -

ENDORSEMENT FOR ADVANCEMENT OF ANNIVERSARY DATE
WHERE REGULATION 9.8 APPLIES

In accordance with regulation 9.8 of Annex VI of the Convention, the new anniversary date is (dd/mm/yyyy)

Signed. .
(signature of duly authorized official)

Place

Date (dd/mm/yyyy) .

(seal or stamp of the authority, as appropriate)

In accordance with regulation 9.8 of Annex VI of the Convention, the new anniversary date is (dd/mm/yyyy)..

Signed..
(signature of duly authorized official)

Place .

Date (dd/mm/yyyy)

(seal or stamp of the authority, as appropriate)

RESOLUTION MEPC.328(76)

Signed..
(signature of duly authorized official)

Place .

Date (dd/mm/yyyy)

(seal or stamp of the authority, as appropriate)

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 52 -

SUPPLEMENT TO
INTERNATIONAL AIR POLLUTION PREVENTION CERTIFICATE (IAPP CERTIFICATE)

RECORD OF CONSTRUCTION AND EQUIPMENT

1 Particulars of ship

1.1 Name of ship.....

1.2 IMO number.....

1.3 Date on which keel was laid or ship was at a similar stage of construction (dd/mm/yyyy)..

1.4 Length (L) metres .....

2 Control of emissions from ships

2.1 Ozone-depleting substances (regulation 12)

2.1.1 The following fire -extinguishing systems, other systems and equipment containing

2 Control of emissions from ships

2.1 Ozone-depleting substances (regulation 12)

2.1.1 The following fire -extinguishing systems, other systems and equipment containing ozone-depleting substances, other than hydrochlorofluorocarbons (HCFCs), installed before 19 May 2005 may continue in service:

System or equipment Location on board Substance

Notes
1 This Record shall be permanently attached to the IAPP Certificate. The IAPP Certificate shall be available on board the ship at all times.
2 The Record shall be at least in English, French or Spanish. If an official language of the issuing country is also used, this shall prevail in case of a dispute or discrepancy.
3 Entries in boxes shall be made by inserting either: a cross (x) for the answers "yes" and "applicable"; or a dash () for the answers "no" and "not applicable", as appropriate.
4 Unless otherwise stated, regulations mentioned in this Record refer to regulations of Annex VI of the Convention and resolutions or circulars refer to those adopted by the International Maritime Organization.
RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE

and resolutions or circulars refer to those adopted by the International Maritime Organization.
RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 53 -

2.1.2 The following systems containing HCFCs installed before 1 January 2020 may continue in service:

System or equipment	Location on board	Substance

2.2 Nitrogen oxides (NOx) (regulation 13)

2.2.1 The following marine diesel engines installed on this ship are in accordance with the requirements of regulation 13, as indicated:

Applicable regulation of MARPOL Annex VI (NTC = NOx Technical Code 2008) (AM = approved method)	Engine #1	Engine #2	Engine #3	Engine #4	Engine #5
1 Manufacturer and model					
2 Serial number					
3 Use (applicable application cycle(s) NTC 3.2)					
4 Rated power (kW) (NTC 1.3.11)					
5 Rated speed (rpm) (NTC 1.3.12)					
6 Identical engine installed 1/1/2000 exempted by 13.1.1.2					
7 Identical engine installation date (dd/mm/yyyy) as per 13.1.1.2					
8a Major conversion					

4 Rated power (kW) (NTC 1.3.11)
5 Rated speed (rpm) (NTC 1.3.12)
6 Identical engine installed 1/1/2000
exempted by 13.1.1.2

7 Identical engine installation date (dd/mm/yyyy)
as per 13.1.1.2

8a
Major conversion
(dd/mm/yyyy)
13.2.1.1 & 13.2.2
8b 13.2.1.2 & 13.2.3
8c 13.2.1.3 & 13.2.3
9a

Tier I
13.3
9b 13.2.2
9c 13.2.3.1
9d 13.2.3.2
9e 13.7.1.2
10a

Tier II
13.4
10b 13.2.2
10c 13.2.2 (Tier III not
possible)

10d 13.2.3.2
10e 13.5.2 (Exemptions)
10f 13.7.1.2
11a
NOx Tier III Emission
Control Areas
13.5.1.1
11b 13.2.2
11c 13.2.3.2
11d 13.7.1.2
RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE
INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS,
1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 54 -

Applicable regulation of MARPOL Annex VI
(NTC = NOx Technical Code 2008)
(AM = approved method)
Engin
e #1
Engin
e #2
Engin
e #3
Engin

e #4
Engin
e #5
12

AM

1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 54 -

Applicable regulation of MARPOL Annex VI
(NTC = NOx Technical Code 2008)
(AM = approved method)
Engin
e #1
Engin
e #2
Engin
e #3
Engin
e #4
Engin
e #5
12

AM
installed
13 not commercially available
at this survey

14 not applicable

2.3 Sulphur oxides (SOx) and particulate matter (regulation 14)

2.3.1 When the ship operates outside of an emission control area specified in regulation 14.3, the ship uses:

.1 fuel oil with a sulphur content as documented by bunker delivery notes that does not exceed the limit value of 0.50% m/m, and/or .....

.2 an equivalent arrangement approved in accordance with regulation 4.1 as listed in paragraph 2.6 that is at least as effective in terms of SOx emission reductions as compared to using a fuel oil with a sulphur content limit value of 0.50% m/m .....

2.3.2 When the ship operates inside an emission control area specified in regulation 14.3, the ship uses:

.1 fuel oil with a sulphur content as documented by bunker delivery notes that

2.3.2 When the ship operates inside an emission control area specified in regulation 14.3, the ship uses:

.1 fuel oil with a sulphur content as documented by bunker delivery notes that does not exceed the limit value of 0.10% m/m,
and/or .

.2 an equivalent arrangement approved in accordance with regulation 4.1 as listed in paragraph 2.6 that is at least as effective in terms of SOx emission reductions as compared to using a fuel oil with a sulphur content limit value of 0.10% m/m ...

2.3.3 For a ship wit hout an equivalent arrangement approved in accordance with regulation 4.1 as listed in paragraph 2.6, the sulphur content of fuel oil carried for use on board the ship shall not exceed 0.50% m/m as documented by bunker delivery notes ....

2.3.4 The ship is fitted with designated sampling point(s) in accordance with regulation 14.10 or 14.11.....

2.3.5 In accordance with regulation 14.12, t he requirement for fitting or designating

regulation 14.10 or 14.11.....

2.3.5 In accordance with regulation 14.12, t he requirement for fitting or designating sampling point(s) in accordance with regulation 14.10 or 14.11 is not applicable for a fuel oil service system for a low-flashpoint fuel for combustion purposes for propulsion or operation on board the ship .....

2.4 Volatile organic compounds (VOCs) (regulation 15)

2.4.1 The tanker has a vapour collection system installed and approved in accordance with MSC/Circ.585.....

2.4.2.1 For a tanker carrying crude oil, there is an approved VOC management plan .....

2.4.2.2 VOC management plan approval reference .....

2.5 Shipboard incineration (regulation 16)

The ship has an incinerator:

.1 installed on or after 1 January 2000 that complies with:

.1 resolution MEPC.76(40), as amended .....

.2 resolution MEPC.244(66) .....

.2 installed before 1 January 2000 that complies with:

.1 resolution MEPC.59(33), as amended .....

.2 resolution MEPC.76(40), as amended .....

2.6 Equivalents (regulation 4)

The ship has been allowed to use the following fitting, material, appliance or apparatus to be fitted in a ship or other procedures, alternative fuel oils, or compliance methods used as an alternative to that required by this Annex:

System or equipment Equivalent used Approval reference

fitted in a ship or other procedures, alternative fuel oils, or compliance methods used as an alternative to that required by this Annex:

System or equipment Equivalent used Approval reference

THIS IS TO CERTIFY that this Record is correct in all respects.

Issued at . . . .
(place of issue of the Record)

Date (dd/mm/yyyy) . . . .
(date of issue) (signature of duly authorized
official issuing the Record)

(seal or stamp of the authority, as appropriate)

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 56 -

Appendix II

Test cycles and weighting factors (regulation 13)

The following test cycles and weighting factors shall be applied for verification of compliance of marine diesel engines with the applicable NOx limit in accordance with regulation 13 of this

Test cycles and weighting factors (regulation 13)

The following test cycles and weighting factors shall be applied for verification of compliance of marine diesel engines with the applicable NOx limit in accordance with regulation 13 of this Annex using the test procedure and calculation method as specified in the revised NO x Technical Code 2008.

.1 For constant -speed marine engines for ship main propulsion, including diesel-electric drive, test cycle E2 shall be applied.

.2 For controllable-pitch propeller sets test cycle E2 shall be applied.

.3 For propeller-law-operated m ain and propeller-law-operated auxiliary engines the test cycle E3 shall be applied.

.4 For constant-speed auxiliary engines test cycle D2 shall be applied.

.5 For variable-speed, variable-load auxiliary engines, not included above, test cycle C1 shall be applied.

Test cycle for constant-speed main propulsion application
(including diesel-electric drive and all controllable-pitch propeller installations)

Test cycle type	E2			
Speed	100%	100%	100%	100%
Power	100%	75%	50%	25%
Weighting factor	0.2	0.5	0.15	0.15

(including diesel-electric drive and all controllable-pitch propeller installations)

Test cycle type
E2
Speed 100% 100% 100% 100%
Power 100% 75% 50% 25%
Weighting factor 0.2 0.5 0.15 0.15

Test cycle for propeller-law-operated main and propeller-law-operated auxiliary engine application

Test cycle type
E3
Speed 100% 91% 80% 63%
Power 100% 75% 50% 25%
Weighting factor 0.2 0.5 0.15 0.15

Test cycle for constant-speed auxiliary engine application

Test cycle type
D2
Speed 100% 100% 100% 100% 100%
Power 100% 75% 50% 25% 10%
Weighting factor 0.05 0.25 0.3 0.3 0.1

Test cycle for variable-speed and variable-load auxiliary engine application

Test cycle type
C1
Speed Rated Intermediate Idle
Torque 100% 75% 50% 10% 100% 75% 50% 0%
Weighting factor 0.15 0.15 0.15 0.1 0.1 0.1 0.1 0.15

In the case of an engine to be certified in accordance with paragraph 5.1.1 of regulation 13, the specific emission at each individual mode point shall not exceed the applicable NO x emission limit value by more than 50% except as follows:

the specific emission at each individual mode point shall not exceed the applicable NO x emission limit value by more than 50% except as follows:

.1 The 10% mode point in the D2 test cycle.
.2 The 10% mode point in the C1 test cycle.
.3 The idle mode point in the C1 test cycle.

Appendix III

Criteria and procedures for the designation of emission control areas
(regulations 13.6 and 14.3)

1 Objectives

1.1 The purpose of this appendix is to provide Parties with the criteria and procedures for formulating and submitting proposals for the designation of emission control areas and to set forth the factors to be considered in the assessment of such proposals by the Organization.

formulating and submitting proposals for the designation of emission control areas and to set forth the factors to be considered in the assessment of such proposals by the Organization.

1.2 Emissions of NOx, SOx and particulate matter from ocean-going ships contribute to ambient concentrations of air pollution in cities and coastal areas around the world. Adverse public health and environmental effects associated with air p ollution include premature mortality, cardiopulmonary disease, lung cancer, chronic respiratory ailments, acidification and eutrophication.

1.3 An emission control area should be considered for adoption by the Organization if supported by a demonstrated need to prevent, reduce and control emissions of NOx or SOx and particulate matter or all three types of emissions (hereinafter emissions) from ships.

2 Process for the designation of emission control areas

2.1 A proposal to the Organization for the designation of an emission control area for NOx or SOx and particulate matter or all three types of emissions may be submitted only by Parties. Where two or more Parties have a common interest in a particular area, they should formulate

or SOx and particulate matter or all three types of emissions may be submitted only by Parties. Where two or more Parties have a common interest in a particular area, they should formulate a coordinated proposal.

2.2 A proposal to designate a given area as an emission control area should be submitted to the Organization in accordance with the rules and procedures established by the Organization.

3 Criteria for designation of an emission control area

3.1 The proposal shall include:

.1 a clear delineation of the proposed area of application, along with a reference chart on which the area is marked;

.2 the type or types of emission(s) that is or are being proposed for control (i.e. NOx or SOx and particulate matter or all three types of emissions);

.3 a description of the human populations and environmental areas at risk from the impacts of ship emissions;

.4 an assessment that emissions from ships operating in the proposed area of application are contributing to ambient concentrations of air pollution or to adverse environmental impacts. Such assessment shall include a description of the impacts of the relevant emissions on human health and the

application are contributing to ambient concentrations of air pollution or to adverse environmental impacts. Such assessment shall include a description of the impacts of the relevant emissions on human health and the environment, such as adverse impacts on terrestrial and aquatic ecosystems, areas of natural productivity, critical habitats, water quality, human health, and areas of cultural and scientific significance, if applicable. The sources of relevant data including methodologies used shall be identified;

.5 relevant information, pertaining to the meteorological conditions in the proposed area of application, to the human populations and environmental areas at risk, in particular prevailing wind patterns, or to topographical, geological, oceanographic, morphological or other conditions that contribute to ambient concentrations of air pollution or adverse environmental impacts;

areas at risk, in particular prevailing wind patterns, or to topographical, geological, oceanographic, morphological or other conditions that contribute to ambient concentrations of air pollution or adverse environmental impacts;

.6 the nature of the ship traffic in the proposed emission control area, including the patterns and density of such traffic;

.7 a description of the control measures taken by the proposing Party or Parties addressing land-based sources of NO x, SO x and particulate matter emissions affecting the human populations and environmental areas at risk that are in place and operating concurrently with the consideration of measures to be adopted in relation to provisions of regulations 13 and 14 of Annex VI; and

.8 the relative costs of reducing emissions from ships when compared with land-based controls, and the economic impacts on shipping engaged in international trade.

3.2 The geographical limits of an emission control area will be based on the relevant criteria outlined above, including emissions and deposition from ships navigating in the proposed area, traffic patterns and density, and wind conditions.

3.2 The geographical limits of an emission control area will be based on the relevant criteria outlined above, including emissions and deposition from ships navigating in the proposed area, traffic patterns and density, and wind conditions.

4 Procedures for the assessment and adoption of emission control areas by the Organization

4.1 The Organization shall consider each proposal submitted to it by a Party or Parties.

4.2 In assessing the proposal, the Organization shall take into account the criteria that are to be included in each proposal for adoption as set forth in section 3 above.

4.3 An emission control area shall be designated by means of an amendment to this Annex, considered, adopted and brought into force in accordance with article 16 of the present Convention.

5 Operation of emission control areas

5.1 Parties that have ships navigatin g in the area are encouraged to bring to the Organization any concerns regarding the operation of the area.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS,

Organization any concerns regarding the operation of the area.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 60 -

Appendix IV

Type approval and operating limits for shipboard incinerators (regulation 16)

1 Shipboard incinerators described in regulation 16.6.1 shall possess an IMO Type Approval Certificate for each incinerator. In order to obtain such certificate, the incinerator shall be designed and built to an approved standard as described in regulation 16.6.1. Each model shall be subject to a specified type approval test operation at the factory or an approved test facility, and under the responsibility of the Administration, using the following standard fuel/waste specification for the type approval test for determining whether the incinerator operates within the limits specified in paragraph 2 of this appendix:

Sludge oil consisting of: 75% sludge oil from heavy fuel oil (HFO);
5% waste lubricating oil; and
20% `emulsified water.

Solid waste consisting of: 50% food waste;

operates within the limits specified in paragraph 2 of this appendix:

Sludge oil consisting of: 75% sludge oil from heavy fuel oil (HFO);
5% waste lubricating oil; and
20% `emulsified water.

Solid waste consisting of: 50% food waste;
50% rubbish containing:
approx. 30% paper,
" 40% cardboard,
" 10% rags,
" 20% plastic.

The mixture will have up to 50% moisture and
7% incombustible solids.

2 Incinerators described in regulation 16.6.1 shall operate within the following limits:

O2 in combustion chamber:

6 -12%
CO in flue gas maximum average:

200 mg/MJ
Soot number maximum average: Bacharach 3 or Ringelmann 1
(20% opacity) (a higher soot number is
acceptable only during very short periods
such as starting up)

Unburned components in ash residues:

Maximum 10% by weight
Combustion chamber flue gas outlet
temperature range:
850 -1200°C

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE
INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS,
1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 61 -

Appendix V

Appendix V

Information to be included in the bunker delivery note (regulation 18.5)

1 Name and IMO number of receiving ship

2 Port

3 Date of commencement of delivery

4 Name, address and telephone number of marine fuel oil supplier

5 Product name(s)

6 Quantity in metric tonnes

7 Density at 15°C (kg/m3)

8 Sulphur content (% m/m)

9 A declaration signed and certified by the fuel oil supplier's representative that the fuel oil supplied is in conformity with regulation 18.3 of this Annex and that the sulphur content of the fuel oil supplied does not exceed:

the limit value given by regulation 14.1 of this Annex;

the limit value given by regulation 14.4 of this Annex; or

the purchaser's specified limit value of _____ (% m/m), as completed by the fuel oil supplier 's representative and on the basis of the purchaser 's notification that the fuel oil:

the purchaser's specified limit value of _____ (% m/m), as completed by the fuel oil supplier 's representative and on the basis of the purchaser 's notification that the fuel oil:

.1 is intended to be used in combinat ion with an equivalent means of compliance in accordance with regulation 4 of this Annex; or

.2 is subject to a relevant exemption for a ship to conduct trials for sulphur oxides emission reduction and control technology research in accordance with regulation 3.2 of this Annex.

The declaration shall be completed by the fuel oil supplier's representative by marking the applicable box(es) with a cross (x).
RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 62 -

Appendix VI

Verification procedures for a MARPOL Annex VI fuel oil sa mple (regulation 18.8.2 or regulation 14.8)

The following relevant verification procedure shall be used to determine whether the fuel oil delivered to, in use or carried for use on board a ship has met the applicable sulphur limit of

regulation 14.8)

The following relevant verification procedure shall be used to determine whether the fuel oil delivered to, in use or carried for use on board a ship has met the applicable sulphur limit of regulation 14 of this Annex.

This appendix refers to the following representative MARPOL Annex VI fuel oil samples:

Part 1 sample of fuel oil delivered in accordance with regulation 18.8.1, hereafter referred to as the "MARPOL delivered sample" as defined in regulation 2.1.22.

Part 2 sample of fuel oil in use, intended to be used or carried for use on board in accordance with regulation 14.8, hereafter referred to as the "in-use sample " as defined in regulation 2.1.16 and "onboard sample" as defined in regulation 2.1.24.

Part 1 MARPOL delivered sample

1 General Requirements

1.1 The representative sample of the fuel oil, which is required by regulation 18.8.1 (the MARPOL delivered sample), shall be used to verify the sulphur content of the fuel oil delivered to a ship.

1.2 A Party, through its competent authority, shall manage the verification procedure.

MARPOL delivered sample), shall be used to verify the sulphur content of the fuel oil delivered to a ship.

1.2 A Party, through its competent authority, shall manage the verification procedure.

1.3 A laboratory undertaking the sulphur testing procedure given in this appendix shall have valid accreditation in respect of the test method to be used.

2 Verification Procedure Part 1

2.1 The MARPOL delivered sample shall be conveyed by the competent authority to the laboratory.

2.2 The laboratory shall:

.1 record the details of the seal number and the sample label on the test record;

.2 record the condition of the seal of the sample as received on the test record; and

.3 reject any sample where the seal has been broken prior to receipt and record that rejection on the test record.

2.3 If the seal of the sample as received has not been broken, the laboratory shall proceed with the verification procedure and shall:

.1 unseal the sample;

.2 ensure that the sample is thoroughly homogenized;

.3 draw two subsamples from the sample; and

.1 unseal the sample;

.2 ensure that the sample is thoroughly homogenized;

.3 draw two subsamples from the sample; and

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 63 -

.4 reseal the sample and record the new reseal details on the test record.

2.4 The two subsamples shall be tested in succession, in accordance with the spe cified test method referred to in regulation 2.1.30 of this Annex. For the purposes of this Part 1 verification procedure, the results of the test analysis shall be referred to as '1A' and '1B':

.1 results 1A and 1B shall be recorded on the test record in accordance with the requirements of the test method; and

.2 if the results of 1A and 1B are within the repeatability (r) of the test method, the results shall be considered valid; or

.3 if the results 1A and 1B are not within the repeatability (r) of the test method, both results shall be rejected and two new subsamples shall be taken by

the results shall be considered valid; or

.3 if the results 1A and 1B are not within the repeatability (r) of the test method, both results shall be rejected and two new subsamples shall be taken by the laboratory and tested. The sample bottle shall be resealed in accordance with paragraph 2.3.4 after the new subsamples have been taken;

.4 in the case of two failures to achieve repeatability between 1A and 1B, the cause of that failure shall be investigated by the laboratory and resolved before further testing of the sample is undertaken. On resolution of that repeatability issue, two new subsamples shall be taken in accordance with paragraph 2.3. The sample shall be resealed in accordance with paragraph 2.3.4 after the new subsamples have been taken.

2.5 If the test results of 1A and 1B are valid, an average of these two results shall be calculated. The average value shall be referred to as 'X' and shall be recorded on the test record:

.1 if the result X is equal to or less than the applicable limit required by regulation 14, the fuel oil shall be considered to have met the requirement; or

record:

.1 if the result X is equal to or less than the applicable limit required by regulation 14, the fuel oil shall be considered to have met the requirement; or

.2 if the result X is greater than the applicable limit required by regulation 14, the fuel oil shall be considered to have not met the requirement.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 64 -

Table 1: Summary of Part 1 MARPOL delivered sample procedure

On the basis of the test method referred to in regulation 2.1.30 of this Annex
Applicable limit % m/m: V Result 2.5.1: X V

Result 2.5.2: X > V
0.10 Met the requirement Not met the requirement
0.50
Result X reported to 2 decimal places

2.6 The final results obtained from this verification procedure shall be evaluated by the competent authority.

2.7 The laboratory shall provide a copy of the test record to the competent authority managing the verification procedure.

Part 2 In-use and onboard samples

3 General Requirements

competent authority.

2.7 The laboratory shall provide a copy of the test record to the competent authority managing the verification procedure.

Part 2 In-use and onboard samples

3 General Requirements

3.1 The in-use or onboard sample, as appropriate, shall be used to verify the sulphur content of the fuel oil as represented by that sample of fuel oil at the point of sampling.

3.2 A Party, through its competent authority, shall manage the verification procedure.

3.3 A laboratory undertaking the sulphur testing procedure given in this appendix shall have valid accreditation in respect of the test method to be used.

4 Verification Procedure Part 2

4.1 The in-use or onboard sample shall be conveyed by the competent authority to the laboratory.

4.2 The laboratory shall:

.1 record the details of the seal number and the sample label on the test record;

.2 record the condition of the seal of the sample as received on the test record; and

.3 reject any sample where the seal has been broken prior to receipt and record that rejection on the test record.

4.3 If the seal of the sample as received has not been broken, the laboratory shall

and

.3 reject any sample where the seal has been broken prior to receipt and record that rejection on the test record.

4.3 If the seal of the sample as received has not been broken, the laboratory shall proceed with the verification procedure and shall:

.1 unseal the sample;

.2 ensure that the sample is thoroughly homogenized;

.3 draw two subsamples from the sample; and

.4 reseal the sample and record the new reseal details on the test record.

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 65 -

4.4 The two subsamples shall be tested in succession, in accordance with the specifi ed test method referred to in regulation 2.1.30 of this Annex. For the purposes of this Part 2 verification procedure, the results obtained shall be referred to as '2A' and '2B':

.1 results 2A and 2B shall be recorded on the test record in accordance with the requirements of the test method; and

.2 if the results of 2A and 2B are within the repeatability (r) of the test method,

.1 results 2A and 2B shall be recorded on the test record in accordance with the requirements of the test method; and

.2 if the results of 2A and 2B are within the repeatability (r) of the test method, the results shall be considered valid; or

.3 if the results of 2A and 2B are not within the repeatability (r) of the test method, both results shall be rejected and two new subsamples shall be taken by the laboratory and tested. The sample bottle shall be resealed in accordance with paragraph 4.3.4 a fter the new subsamples have been taken; and

.4 in the case of two failures to achieve repeatability between 2A and 2B, the cause of that failure shall be investigated by the laboratory and resolved before further testing of the sample is undertaken. On resolution of that repeatability issue, two new subsamples shall be taken in accordance with paragraph 4.3. The sample shall be resealed in accordance with paragraph 4.3.4 after the new subsamples have been taken.

4.5 If the test results of 2A and 2B are valid, an average of these two results shall be calculated. That average value shall be referred to as 'Z' and shall be recorded on the test record:

4.3.4 after the new subsamples have been taken.

4.5 If the test results of 2A and 2B are valid, an average of these two results shall be calculated. That average value shall be referred to as 'Z' and shall be recorded on the test record:

.1 if Z is equal to or less than the applicable limit required by regulation 14, the sulphur content of the fuel oil as represented by the tested sample shall be considered to have met the requirement;

.2 if Z is greater than the applicable limit required by regulation 14 but less than or equal to that applicable limit + 0.59R (where R is the reproducibility of the test method), the sulphur content of the fuel oil as represented by the tested sample shall be considered to have met the requirement; or

.3 if Z is greater than the applicable limit required by regulation 14 + 0.59R, the sulphur content of the fuel oil as represented by the tested sample shall be considered to have not met the requirement.

Table 2: Summary of in-use or onboard sample procedure

On the basis of the test method referred to in regulation 2.1.30 of this Annex				
Applicable limit %m/m:				
V				
Test margin value:				
W				
Result 4.5.1:				
Z V				
Result 4.5.2:				
V < Z W				
Result 4.5.3:				
Z > W				
0.10	0.11	Met the requirement		
			Met the requirement	
				Not met the requirement
0.50	0.53			

Result Z reported to 2 decimal places

4.6 The final results obtained from this verification procedure shall be evaluated by the competent authority.

4.7 The laboratory shall provide a copy of the test record to the competent authority managing the verification procedure.

Appendix VII

Emission control areas (regulations 13.6 and 14.3)

Appendix VII

Emission control areas (regulations 13.6 and 14.3)

1 The boundaries of emission control areas designated under regulations 13.6 and 14.3, other than the Baltic Sea and the North Sea areas, are set forth in this appendix.

2 The North American area comprises:

.1 the sea area located off the Pacific coasts of the United States and Canada, enclosed by geodesic lines connecting the following coordinates:

Point	Latitude	Longitude
1	32°32.10 N	117°06.11 W
2	32°32.04 N	117°07.29 W
3	32°31.39 N	117°14.20 W
4	32°33.13 N	117°15.50 W
5	32°34.21 N	117°22.01 W
6	32°35.23 N	117°27.53 W
7	32°37.38 N	117°49.34 W
8	31°07.59 N	118°36.21 W
9	30°33.25 N	121°47.29 W
10	31°46.11 N	123°17.22 W
11	32°21.58 N	123°50.44 W
12	32°56.39 N	124°11.47 W
13	33°40.12 N	124°27.15 W
14	34°31.28 N	125°16.52 W
15	35°14.38 N	125°43.23 W
16	35°44.00 N	126°18.53 W
17	36°16.25 N	126°45.30 W
18	37°01.35 N	127°07.18 W
19	37°45.39 N	127°38.02 W
20	38°25.08 N	127°53.00 W
21	39°25.05 N	128°31.23 W

14 34°31.28 N 125°16.52 W
15 35°14.38 N 125°43.23 W
16 35°44.00 N 126°18.53 W
17 36°16.25 N 126°45.30 W
18 37°01.35 N 127°07.18 W
19 37°45.39 N 127°38.02 W
20 38°25.08 N 127°53.00 W
21 39°25.05 N 128°31.23 W
22 40°18.47 N 128°45.46 W
23 41°13.39 N 128°40.22 W
24 42°12.49 N 129°00.38 W
25 42°47.34 N 129°05.42 W
26 43°26.22 N 129°01.26 W
27 44°24.43 N 128°41.23 W
28 45°30.43 N 128°40.02 W
29 46°11.01 N 128°49.01 W
30 46°33.55 N 129°04.29 W
31 47°39.55 N 131°15.41 W
32 48°32.32 N 132°41.00 W
33 48°57.47 N 133°14.47 W

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 68 -

Point Latitude Longitude
34 49°22.39 N 134°15.51 W
35 50°01.52 N 135°19.01 W
36 51°03.18 N 136°45.45 W
37 51°54.04 N 137°41.54 W
38 52°45.12 N 138°20.14 W
39 53°29.20 N 138°40.36 W
40 53°40.39 N 138°48.53 W
41 54°13.45 N 139°32.38 W
42 54°39.25 N 139°56.19 W
43 55°20.18 N 140°55.45 W
44 56°07.12 N 141°36.18 W
45 56°28.32 N 142°17.19 W

38 52°45.12 N 138°20.14 W
39 53°29.20 N 138°40.36 W
40 53°40.39 N 138°48.53 W
41 54°13.45 N 139°32.38 W
42 54°39.25 N 139°56.19 W
43 55°20.18 N 140°55.45 W
44 56°07.12 N 141°36.18 W
45 56°28.32 N 142°17.19 W
46 56°37.19 N 142°48.57 W
47 58°51.04 N 153°15.03 W

.2 the sea areas located off the Atlantic coasts of the United States, Canada and France (Saint- Pierre-et-Miquelon), and the Gulf of Mexico coast of the United States enclosed by geodesic lines connecting the following coordinates:

Point Latitude Longitude
1 60°00.00 N 64°09.36 W
2 60°00.00 N 56°43.00 W
3 58°54.01 N 55°38.05 W
4 57°50.52 N 55°03.47 W
5 57°35.13 N 54°00.59 W
6 57°14.20 N 53°07.58 W
7 56°48.09 N 52°23.29 W
8 56°18.13 N 51°49.42 W
9 54°23.21 N 50°17.44 W
10 53°44.54 N 50°07.17 W
11 53°04.59 N 50°10.05 W
12 52°20.06 N 49°57.09 W
13 51°34.20 N 48°52.45 W
14 50°40.15 N 48°16.04 W
15 50°02.28 N 48°07.03 W
16 49°24.03 N 48°09.35 W
17 48°39.22 N 47°55.17 W
18 47°24.25 N 47°46.56 W
19 46°35.12 N 48°00.54 W
20 45°19.45 N 48°43.28 W
21 44°43.38 N 49°16.50 W
22 44°16.38 N 49°51.23 W
RESOLUTION MEPC.328(76)

15 50°02.28 N 48°07.03 W
16 49°24.03 N 48°09.35 W
17 48°39.22 N 47°55.17 W
18 47°24.25 N 47°46.56 W
19 46°35.12 N 48°00.54 W
20 45°19.45 N 48°43.28 W
21 44°43.38 N 49°16.50 W
22 44°16.38 N 49°51.23 W
RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 69 -

Point Latitude Longitude
23 43°53.15 N 50°34.01 W
24 43°36.06 N 51°20.41 W
25 43°23.59 N 52°17.22 W
26 43°19.50 N 53°20.13 W
27 43°21.14 N 54°09.20 W
28 43°29.41 N 55°07.41 W
29 42°40.12 N 55°31.44 W
30 41°58.19 N 56°09.34 W
31 41°20.21 N 57°05.13 W
32 40°55.34 N 58°02.55 W
33 40°41.38 N 59°05.18 W
34 40°38.33 N 60°12.20 W
35 40°45.46 N 61°14.03 W
36 41°04.52 N 62°17.49 W
37 40°36.55 N 63°10.49 W
38 40°17.32 N 64°08.37 W
39 40°07.46 N 64°59.31 W
40 40°05.44 N 65°53.07 W
41 39°58.05 N 65°59.51 W
42 39°28.24 N 66°21.14 W
43 39°01.54 N 66°48.33 W
44 38°39.16 N 67°20.59 W
45 38°19.20 N 68°02.01 W
46 38°05.29 N 68°46.55 W
47 37°58.14 N 69°34.07 W

40 40°05.44 N 65°53.07 W
41 39°58.05 N 65°59.51 W
42 39°28.24 N 66°21.14 W
43 39°01.54 N 66°48.33 W
44 38°39.16 N 67°20.59 W
45 38°19.20 N 68°02.01 W
46 38°05.29 N 68°46.55 W
47 37°58.14 N 69°34.07 W
48 37°57.47 N 70°24.09 W
49 37°52.46 N 70°37.50 W
50 37°18.37 N 71°08.33 W
51 36°32.25 N 71°33.59 W
52 35°34.58 N 71°26.02 W
53 34°33.10 N 71°37.04 W
54 33°54.49 N 71°52.35 W
55 33°19.23 N 72°17.12 W
56 32°45.31 N 72°54.05 W
57 31°55.13 N 74°12.02 W
58 31°27.14 N 75°15.20 W
59 31°03.16 N 75°51.18 W
60 30°45.42 N 76°31.38 W
61 30°12.48 N 77°18.29 W
62 29°25.17 N 76°56.42 W
63 28°36.59 N 76°48.00 W
64 28°17.13 N 76°40.10 W

Point Latitude Longitude
65 28°17.12 N 79°11.23 W
66 27°52.56 N 79°28.35 W
67 27°26.01 N 79°31.38 W
68 27°16.13 N 79°34.18 W
69 27°11.54 N 79°34.56 W
70 27°05.59 N 79°35.19 W
71 27°00.28 N 79°35.17 W
72 26°55.16 N 79°34.39 W

Point	Latitude	Longitude
65	28°17.12 N	79°11.23 W
66	27°52.56 N	79°28.35 W
67	27°26.01 N	79°31.38 W
68	27°16.13 N	79°34.18 W
69	27°11.54 N	79°34.56 W
70	27°05.59 N	79°35.19 W
71	27°00.28 N	79°35.17 W
72	26°55.16 N	79°34.39 W
73	26°53.58 N	79°34.27 W
74	26°45.46 N	79°32.41 W
75	26°44.30 N	79°32.23 W
76	26°43.40 N	79°32.20 W
77	26°41.12 N	79°32.01 W
78	26°38.13 N	79°31.32 W
79	26°36.30 N	79°31.06 W
80	26°35.21 N	79°30.50 W
81	26°34.51 N	79°30.46 W
82	26°34.11 N	79°30.38 W
83	26°31.12 N	79°30.15 W
84	26°29.05 N	79°29.53 W
85	26°25.31 N	79°29.58 W
86	26°23.29 N	79°29.55 W
87	26°23.21 N	79°29.54 W
88	26°18.57 N	79°31.55 W
89	26°15.26 N	79°33.17 W
90	26°15.13 N	79°33.23 W
91	26°08.09 N	79°35.53 W
92	26°07.47 N	79°36.09 W
93	26°06.59 N	79°36.35 W
94	26°02.52 N	79°38.22 W
95	25°59.30 N	79°40.03 W
96	25°59.16 N	79°40.08 W
97	25°57.48 N	79°40.38 W
98	25°56.18 N	79°41.06 W
99	25°54.04 N	79°41.38 W
100	25°53.24 N	79°41.46 W
101	25°51.54 N	79°41.59 W
102	25°49.33 N	79°42.16 W
103	25°48.24 N	79°42.23 W
104	25°48.20 N	79°42.24 W
105	25°46.26 N	79°42.44 W

98 25°56.18 N 79°41.06 W
99 25°54.04 N 79°41.38 W
100 25°53.24 N 79°41.46 W
101 25°51.54 N 79°41.59 W
102 25°49.33 N 79°42.16 W
103 25°48.24 N 79°42.23 W
104 25°48.20 N 79°42.24 W
105 25°46.26 N 79°42.44 W
106 25°46.16 N 79°42.45 W
RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 71 -

Point Latitude Longitude
107 25°43.40 N 79°42.59 W
108 25°42.31 N 79°42.48 W
109 25°40.37 N 79°42.27 W
110 25°37.24 N 79°42.27 W
111 25°37.08 N 79°42.27 W
112 25°31.03 N 79°42.12 W
113 25°27.59 N 79°42.11 W
114 25°24.04 N 79°42.12 W
115 25°22.21 N 79°42.20 W
116 25°21.29 N 79°42.08 W
117 25°16.52 N 79°41.24 W
118 25°15.57 N 79°41.31 W
119 25°10.39 N 79°41.31 W
120 25°09.51 N 79°41.36 W
121 25°09.03 N 79°41.45 W
122 25°03.55 N 79°42.29 W
123 25°03.00 N 79°42.56 W
124 25°00.30 N 79°44.05 W
125 24°59.03 N 79°44.48 W
126 24°55.28 N 79°45.57 W
127 24°44.18 N 79°49.24 W
128 24°43.04 N 79°49.38 W
129 24°42.36 N 79°50.50 W

122 25°03.55 N 79°42.29 W
123 25°03.00 N 79°42.56 W
124 25°00.30 N 79°44.05 W
125 24°59.03 N 79°44.48 W
126 24°55.28 N 79°45.57 W
127 24°44.18 N 79°49.24 W
128 24°43.04 N 79°49.38 W
129 24°42.36 N 79°50.50 W
130 24°41.47 N 79°52.57 W
131 24°38.32 N 79°59.58 W
132 24°36.27 N 80°03.51 W
133 24°33.18 N 80°12.43 W
134 24°33.05 N 80°13.21 W
135 24°32.13 N 80°15.16 W
136 24°31.27 N 80°16.55 W
137 24°30.57 N 80°17.47 W
138 24°30.14 N 80°19.21 W
139 24°30.06 N 80°19.44 W
140 24°29.38 N 80°21.05 W
141 24°28.18 N 80°24.35 W
142 24°28.06 N 80°25.10 W
143 24°27.23 N 80°27.20 W
144 24°26.30 N 80°29.30 W
145 24°25.07 N 80°32.22 W
146 24°23.30 N 80°36.09 W
147 24°22.33 N 80°38.56 W
148 24°22.07 N 80°39.51 W

Point Latitude Longitude
149 24°19.31 N 80°45.21 W
150 24°19.16 N 80°45.47 W
151 24°18.38 N 80°46.49 W
152 24°18.35 N 80°46.54 W
153 24°09.51 N 80°59.47 W

Point	Latitude	Longitude
149	24°19.31 N	80°45.21 W
150	24°19.16 N	80°45.47 W
151	24°18.38 N	80°46.49 W
152	24°18.35 N	80°46.54 W
153	24°09.51 N	80°59.47 W
154	24°09.48 N	80°59.51 W
155	24°08.58 N	81°01.07 W
156	24°08.30 N	81°01.51 W
157	24°08.26 N	81°01.57 W
158	24°07.28 N	81°03.06 W
159	24°02.20 N	81°09.05 W
160	24°00.00 N	81°11.16 W
161	23°55.32 N	81°12.55 W
162	23°53.52 N	81°19.43 W
163	23°50.52 N	81°29.59 W
164	23°50.02 N	81°39.59 W
165	23°49.05 N	81°49.59 W
166	23°49.05 N	82°00.11 W
167	23°49.42 N	82°09.59 W
168	23°51.14 N	82°24.59 W
169	23°51.14 N	82°39.59 W
170	23°49.42 N	82°48.53 W
171	23°49.32 N	82°51.11 W
172	23°49.24 N	82°59.59 W
173	23°49.52 N	83°14.59 W
174	23°51.22 N	83°25.49 W
175	23°52.27 N	83°33.01 W
176	23°54.04 N	83°41.35 W
177	23°55.47 N	83°48.11 W
178	23°58.38 N	83°59.59 W
179	24°09.37 N	84°29.27 W
180	24°13.20 N	84°38.39 W
181	24°16.41 N	84°46.07 W
182	24°23.30 N	84°59.59 W
183	24°26.37 N	85°06.19 W
184	24°38.57 N	85°31.54 W
185	24°44.17 N	85°43.11 W
186	24°53.57 N	85°59.59 W

179 24°09.37 N 84°29.27 W
180 24°13.20 N 84°38.39 W
181 24°16.41 N 84°46.07 W
182 24°23.30 N 84°59.59 W
183 24°26.37 N 85°06.19 W
184 24°38.57 N 85°31.54 W
185 24°44.17 N 85°43.11 W
186 24°53.57 N 85°59.59 W
187 25°10.44 N 86°30.07 W
188 25°43.15 N 86°21.14 W
189 26°13.13 N 86°06.45 W
190 26°27.22 N 86°13.15 W
RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 73 -

Point Latitude Longitude
191 26°33.46 N 86°37.07 W
192 26°01.24 N 87°29.35 W
193 25°42.25 N 88°33.00 W
194 25°46.54 N 90°29.41 W
195 25°44.39 N 90°47.05 W
196 25°51.43 N 91°52.50 W
197 26°17.44 N 93°03.59 W
198 25°59.55 N 93°33.52 W
199 26°00.32 N 95°39.27 W
200 26°00.33 N 96°48.30 W
201 25°58.32 N 96°55.28 W
202 25°58.15 N 96°58.41 W
203 25°57.58 N 97°01.54 W
204 25°57.41 N 97°05.08 W
205 25°57.24 N 97°08.21 W
206 25°57.24 N 97°08.47 W

.3 the sea area located off the coasts of the Hawaiian Islands of Hawai'i, Maui,

202 25°58.15 N 96°58.41 W
203 25°57.58 N 97°01.54 W
204 25°57.41 N 97°05.08 W
205 25°57.24 N 97°08.21 W
206 25°57.24 N 97°08.47 W

.3 the sea area located off the coasts of the Hawaiian Islands of Hawai'i, Maui, Oahu, Moloka 'i, Ni 'ihau, Kaua 'i, Lana 'i and Kaho'olawe, enclosed by geodesic lines connecting the following coordinates:

Point Latitude Longitude
1 22°32.54 N 153°00.33 W
2 23°06.05 N 153°28.36 W
3 23°32.11 N 154°02.12 W
4 23°51.47 N 154°36.48 W
5 24°21.49 N 155°51.13 W
6 24°41.47 N 156°27.27 W
7 24°57.33 N 157°22.17 W
8 25°13.41 N 157°54.13 W
9 25°25.31 N 158°30.36 W
10 25°31.19 N 159°09.47 W
11 25°30.31 N 159°54.21 W
12 25°21.53 N 160°39.53 W
13 25°00.06 N 161°38.33 W
14 24°40.49 N 162°13.13 W
15 24°15.53 N 162°43.08 W
16 23°40.50 N 163°13.00 W
17 23°03.20 N 163°32.58 W
18 22°20.09 N 163°44.41 W
19 21°36.45 N 163°46.03 W
20 20°55.26 N 163°37.44 W
21 20°13.34 N 163°19.13 W

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS,

19 21°36.45 N 163°46.03 W
20 20°55.26 N 163°37.44 W
21 20°13.34 N 163°19.13 W
RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 74 -

Point Latitude Longitude
22 19°39.03 N 162°53.48 W
23 19°09.43 N 162°20.35 W
24 18°39.16 N 161°19.14 W
25 18°30.31 N 160°38.30 W
26 18°29.31 N 159°56.17 W
27 18°10.41 N 159°14.08 W
28 17°31.17 N 158°56.55 W
29 16°54.06 N 158°30.29 W
30 16°25.49 N 157°59.25 W
31 15°59.57 N 157°17.35 W
32 15°40.37 N 156°21.06 W
33 15°37.36 N 155°22.16 W
34 15°43.46 N 154°46.37 W
35 15°55.32 N 154°13.05 W
36 16°46.27 N 152°49.11 W
37 17°33.42 N 152°00.32 W
38 18°30.16 N 151°30.24 W
39 19°02.47 N 151°22.17 W
40 19°34.46 N 151°19.47 W
41 20°07.42 N 151°22.58 W
42 20°38.43 N 151°31.36 W
43 21°29.09 N 151°59.50 W
44 22°06.58 N 152°31.25 W
45 22°32.54 N 153°00.33 W

3 The United States Caribbean Sea area includes:

.1 the sea area located off the Atlantic and Caribbean coasts of the

42 20°38.43 N 151°31.36 W
43 21°29.09 N 151°59.50 W
44 22°06.58 N 152°31.25 W
45 22°32.54 N 153°00.33 W

3 The United States Caribbean Sea area includes:

.1 the sea area located off the Atlantic and Caribbean coasts of the Commonwealth of Puerto Rico and the United States Virgin Islands, enclosed by geodesic lines connecting the following coordinates:

Point Latitude Longitude
1 17°18.37 N 67°32.14 W
2 19°11.14 N 67°26.45 W
3 19°30.28 N 65°16.48 W
4 19°12.25 N 65°06.08 W
5 18°45.13 N 65°00.22 W
6 18°41.14 N 64°59.33 W
7 18°29.22 N 64°53.51 W
8 18°27.35 N 64°53.22 W
9 18°25.21 N 64°52.39 W
10 18°24.30 N 64°52.19 W
11 18°23.51 N 64°51.50 W
12 18°23.42 N 64°51.23 W

Point Latitude Longitude
13 18°23.36 N 64°50.17 W
14 18°23.48 N 64°49.41 W
15 18°24.11 N 64°49.00 W
16 18°24.28 N 64°47.57 W
17 18°24.18 N 64°47.01 W
18 18°23.13 N 64°46.37 W
19 18°22.37 N 64°45.20 W

Point	Latitude	Longitude
13	18°23.36 N	64°50.17 W
14	18°23.48 N	64°49.41 W
15	18°24.11 N	64°49.00 W
16	18°24.28 N	64°47.57 W
17	18°24.18 N	64°47.01 W
18	18°23.13 N	64°46.37 W
19	18°22.37 N	64°45.20 W
20	18°22.39 N	64°44.42 W
21	18°22.42 N	64°44.36 W
22	18°22.37 N	64°44.24 W
23	18°22.39 N	64°43.42 W
24	18°22.30 N	64°43.36 W
25	18°22.25 N	64°42.58 W
26	18°22.26 N	64°42.28 W
27	18°22.15 N	64°42.03 W
28	18°22.22 N	64°40.60 W
29	18°21.57 N	64°40.15 W
30	18°21.51 N	64°38.23 W
31	18°21.22 N	64°38.16 W
32	18°20.39 N	64°38.33 W
33	18°19.15 N	64°38.14 W
34	18°19.07 N	64°38.16 W
35	18°17.23 N	64°39.38 W
36	18°16.43 N	64°39.41 W
37	18°11.33 N	64°38.58 W
38	18°03.02 N	64°38.03 W
39	18°02.56 N	64°29.35 W
40	18°02.51 N	64°27.02 W
41	18°02.30 N	64°21.08 W
42	18°02.31 N	64°20.08 W
43	18°02.03 N	64°15.57 W
44	18°00.12 N	64°02.29 W
45	17°59.58 N	64°01.04 W
46	17°58.47 N	63°57.01 W
47	17°57.51 N	63°53.54 W
48	17°56.38 N	63°53.21 W
49	17°39.40 N	63°54.53 W
50	17°37.08 N	63°55.10 W
51	17°30.21 N	63°55.56 W
52	17°11.36 N	63°57.57 W
53	17°05.00 N	63°58.41 W

46 17°58.47 N 63°57.01 W
47 17°57.51 N 63°53.54 W
48 17°56.38 N 63°53.21 W
49 17°39.40 N 63°54.53 W
50 17°37.08 N 63°55.10 W
51 17°30.21 N 63°55.56 W
52 17°11.36 N 63°57.57 W
53 17°05.00 N 63°58.41 W
54 16°59.49 N 63°59.18 W
55 17°18.37 N 67°32.14 W

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 76 -

Appendix VIII

Form of International Energy Efficiency (IEE) Certificate (regulation 8.2)

INTERNATIONAL ENERGY EFFICIENCY CERTIFICATE

Issued under the provisions of the Protocol of 1997, as amended, to amend the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto (hereinafter referred to as "the Convention") under the authority of the Government of:

.....
(full designation of the country)

Government of:

.....
(full designation of the country)

by .....
(full designation of the competent person or organization
authorized under the provisions of the Convention)

Particulars of ship

Name of ship .....

Distinctive number or letters .....

Port of registry .....

Gross tonnage .....

IMO number .....

IMO number .....

THIS IS TO CERTIFY:

1 That the ship has been surveyed in accordance with regulation 5.4 of Annex VI to the Convention; and

2 That the survey shows that the ship complies with the applicable requirements in regulations 22, 23, 24, 25 and 26.

Completion date of survey on which this Certificate is based: ..... (dd/mm/yyyy)

Issued at .....
(place of issue of certificate)

(dd/mm/yyyy): ....
(date of issue) (signature of duly authorized official
issuing the certificate)
(seal or stamp of the authority, as appropriate)

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 77 -

Supplement to the International Energy Efficiency Certificate
(IEE Certificate)

RECORD OF CONSTRUCTION RELATING TO ENERGY EFFICIENCY

Notes:

1 This Record shall be permanently attached to the IEE Certificate. The IEE Certificate shall be available on board the ship at all times.

2 The Record shall be at least in English, French or Spanish. If an official language of the issuing Party is also used, this shall prevail in case of a dispute or discrepancy.

3 Entries in boxes shall be made by inserting either: a cross (x) for the answers "yes" and "applicable"; or a dash (-) for the answers "no" and "not applicable ", as appropriate.

4 Unless otherwise stated, regulations mentioned in this Record refer to regulations in Annex VI of the Convention, and resolutions or circulars refer to those adopted by the International Maritime Organization.

1 Particulars of ship

4 Unless otherwise stated, regulations mentioned in this Record refer to regulations in Annex VI of the Convention, and resolutions or circulars refer to those adopted by the International Maritime Organization.

1 Particulars of ship

1.1 Name of ship .....

1.2 IMO number .....

1.3 Date of building contract .....

1.4 Date of major conversion (if applicable)

1.5 Gross tonnage .....

1.6 Deadweight .....

1.7 Type of ship .....

2 Propulsion system

1.7 Type of ship .....

2 Propulsion system

2.1 Diesel propulsion .....

2.2 Diesel-electric propulsion .....

2.3 Turbine propulsion .....

2.4 Hybrid propulsion .....

2.5 Propulsion system other than any of the above .....

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 78 -

3 Attained Energy Efficiency Design Index (EEDI)

3.1 The attained EEDI in accordance with regulation 22.1 is calculated based on the information contained in the EEDI technical file, which also shows the process of

3 Attained Energy Efficiency Design Index (EEDI)

3.1 The attained EEDI in accordance with regulation 22.1 is calculated based on the information contained in the EEDI technical file, which also shows the process of calculating the attained EEDI.....

The attained EEDI is: ..... grams-CO2/tonne-nautical mile

3.2 The attained EEDI is not calculated, as:

3.2.1 the ship is exempt under regulation 22.1 as it is not a new ship as defined in regulation 2.2.18 .....

3.2.2 the type of propulsion system is exempt in accordance with regulation 19.3 .....

3.2.3 the requirement of regulation 22 is waived by the ship's Administration in accordance with regulation 19.4 .....

3.2.4 the type of ship is exempt in accordance with regulation 22.1 .....

4 Required EEDI

4.1 Required EEDI is: grams-CO2/tonne-mile

3.2.4 the type of ship is exempt in accordance with regulation 22.1 .....

4 Required EEDI

4.1 Required EEDI is: grams-CO2/tonne-mile

4.2 The required EEDI is not applicable, as:

4.2.1 the ship is exempt under regulation 24.1 as it is not a new ship as defined in regulation 2.2.18.....

4.2.2 the type of propulsion system is exempt in accordance with regulation 19.3.....

4.2.3 the requirement of regulation 24 is waived by the ship's Administration in accordance with regulation 19.4 .....

4.2.4 the type of ship is exempt in accordance with regulation 24.1 .....

4.2.5 the ship 's capacity is below the minimum capacity threshold in table 1 of regulation 24.2.....

5 Attained Energy Efficiency Existing Ship Index (EEXI)

regulation 24.2.....

5 Attained Energy Efficiency Existing Ship Index (EEXI)

5.1 The attained EEXI in accordance with regulation 23.1is calculated taking into account the guidelines developed by the Organization..

The attained EEXI is:..grams-CO2/tonne-mile

5.2 The attained EEXI is not calculated, as:

5.2.1 the type of propulsion system is exempt in accordance with regulation 19.3.

5.2.2 the type of ship is exempt in accordance with regulation 23.1

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 79 -

6 Required EEXI

6.1 The required EEXI is:.grams-CO2/tonne-mile in accordance with regulation 25

6.2 The required EEXI is not applicable, as:

6.2.1 the type of propulsion system is exempt in accordance with regulation 19.3..

6.2.2 the type of ship is exempt in accordance with regulation 25.1

6.2 The required EEXI is not applicable, as:

6.2.1 the type of propulsion system is exempt in accordance with regulation 19.3..

6.2.2 the type of ship is exempt in accordance with regulation 25.1

6.2.3 the ship 's capacity is below the minimum capacity threshold in table 3 of regulation 25.1.

7 Ship Energy Efficiency Management Plan

7.1 The ship is provided with a Ship Energy Efficiency Management Plan (SEEMP) in compliance with regulation 26.....

8 EEDI technical file

8.1 The IEE Certificate is accompanied by the EEDI technical file in compliance with regulation 22.1....

8.1.1 The EEDI technical file identification/verification number..

8.1.2 The EEDI technical file verification date.....

9 EEXI technical file

9.1 The IEE Certificate is ac companied by the EEXI technical file in compliance with regulation 23.1

9.1.1 The EEXI technical file identification/verification number..

9.1.2 The EEXI technical file verification date.....

regulation 23.1

9.1.1 The EEXI technical file identification/verification number..

9.1.2 The EEXI technical file verification date.....

9.2 The IEE Certificate is not accompanied by the EEXI technical file as the attained EEDI is used as an alternative to the attained EEXI..

THIS IS TO CERTIFY that this Record is correct in all respects.

Issued at .....
(place of issue of the Record)

(dd/mm/yyyy): ....
(date of issue) (signature of duly authorized official
issuing the Record)

(seal or stamp of the authority, as appropriate)
RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 80 -

Appendix IX

Information to be submitted to the IMO Ship Fuel Oil Consumption Database
(regulation 27)

Appendix IX

Information to be submitted to the IMO Ship Fuel Oil Consumption Database
(regulation 27)

Identity of the ship

IMO Number . . . . . .

Period of calendar year for which the data is submitted

Start date (dd/mm/yyyy)
.
End date (dd/mm/yyyy)..

Technical characteristics of the ship

Ship type, as defined in regulation 2 of this Annex or other (to be stated)

Gross tonnage (GT) .

Net tonnage (NT) ..

Deadweight tonnage (DWT)

Power output (rated power) of main and auxiliary reciprocating internal combustion engines over 130 kW (to be stated in kW)

EEDI (if applicable)..

Ice class .

over 130 kW (to be stated in kW)

EEDI (if applicable)..

Ice class .

Fuel oil consumption, by fuel oil type in metric tonnes and methods used for collecting fuel oil consumption data

Distance travelled ..

Hours under way..
RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 81 -

Appendix X

Form of Statement of Compliance Fuel Oil Consumption Reporting and Operational Carbon Intensity rating (regulation 8.3)

STATEMENT OF COMPLIANCE FUEL OIL CONSUMPTION REPORTING AND OPERATIONAL CARBON INTENSITY RATING

Issued under the provisions of the Protocol of 1997, as amended, to amend the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto (hereinafter referred to as "the Convention") under the authority of the Government of:

Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto (hereinafter referred to as "the Convention") under the authority of the Government of:

(full designation of the country)

by.....
(full designation of the competent person or organization authorized under the provisions of the Convention)

Particulars of ship

Name of ship

Distinctive number or letters..

IMO number.

Port of registry..

Gross tonnage...

Deadweight.

Type of ship

THIS IS TO DECLARE THAT:

1 the ship has submitted to this Administration the data required by regulation 27 of Annex VI to the Convention, covering ship operations from (dd/mm/yyyy) to (dd/mm/yyyy);

THIS IS TO DECLARE THAT:

1 the ship has submitted to this Administration the data required by regulation 27 of Annex VI to the Convention, covering ship operations from (dd/mm/yyyy) to (dd/mm/yyyy);

2 the data was collected and reported in accordance with the methodology and processes set out in the ship 's SEEMP that was in effect over the period from (dd/mm/yyyy) to (dd/mm/yyyy);

3 the attained annual operational CII of the ship from (dd/mm/yyyy) through (dd/mm/yyyy) was: pursuant to regulations 28.1 and 28.2 of Annex VI of the Convention, for ships to which regulation 28 applies;

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 82 -

4 the annual operational carbon intensity of the ship in this period is rated as
A B C D E

in accordance with regulation 28 of Annex VI to the Convention, for a ship to which regulation 28 applies; and

5 a corrective action plan has been developed and included in the SEEMP (for a ship

A B C D E

in accordance with regulation 28 of Annex VI to the Convention, for a ship to which regulation 28 applies; and

5 a corrective action plan has been developed and included in the SEEMP (for a ship to which regulation 28 applies, rated as D for three consecutive years or rated as E)

This Statement of Compliance is valid until (dd/mm/yyyy) .....

Issued at...
(place of issue of the Statement)

(dd/mm/yyyy): ....
(date of issue) (signature of duly authorized official
issuing the Statement)

(seal or stamp of the authority, as appropriate)

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 83 -

Appendix XI

Form of Exemption Certificate for UNSP Barges (regulation 8.4)

INTERNATIONAL AIR POLLUTION PREVENTION EXEMPTION CERTIFICATE FOR

Appendix XI

Form of Exemption Certificate for UNSP Barges (regulation 8.4)

INTERNATIONAL AIR POLLUTION PREVENTION EXEMPTION CERTIFICATE FOR UNMANNED NON-SELF-PROPELLED (UNSP) BARGES

Issued under the provisions of the Protocol of 1997, as amended, to amend the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto (hereinafter referred to as "the Convention") under the authority of the Government of:

(full designation of the country)

by
(full designation of the competent person or organization
authorized under the provisions of the Convention)

Particulars of ship

Name of ship

Distinctive number or letters..

IMO Number.

Port of registry..

Gross tonnage....

THIS IS TO CERTIFY THAT:

1 the UNSP barge has been surveyed in accordance with regulation 3.4 of Annex VI to

Port of registry..

Gross tonnage....

THIS IS TO CERTIFY THAT:

1 the UNSP barge has been surveyed in accordance with regulation 3.4 of Annex VI to the Convention;

2 the survey shows that the UNSP barge:

.1 is not propelled by mechanical means;

.2 has no system, equipment and/or machinery fitted that may generate emissions controlled by Annex VI to the Convention; and

.3 has neither persons nor living animals on board; and

3 the UNSP barge is exempted, under regulation 3.4 of Annex VI to the Convention from the certification and related survey requirements of regulations 5.1 and 6.1 of Annex VI to the Convention.

This Certificate is valid until (dd/mm/yyyy) .

subject to the exemption conditions being maintained.

Completion date of the survey on which this Certificate is based (dd/mm/yyyy)

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS, 1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO- 84 -

Issued at

Issued at
(place of issue of certificate)

(dd/mm/yyyy): ....
(date of issue) (signature of duly authorized official
issuing the certificate)

(seal or stamp of the authority, as appropriate)

_____

RESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE
INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS,
1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETORESOLUTION MEPC.328(76)
AMENDMENTS TO THE ANNEX OF THE PROTOCOL OF 1997 TO AMEND THE
INTERNATIONAL CONVENTION FOR THE PREVENTION OF POLLUTION FROM SHIPS,
1973, AS MODIFIED BY THE PROTOCOL OF 1978 RELATING THERETO